

We've tried our level best to make this guide as helpful and comprehensive as possible—if you have any suggestions, corrections, or feedback, please feel free to reach out anytime!

B.E. Computer Science **ULTIMATE** Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Logic in Computer Science (CS F214) Course

Content

Logic in Computer Science (LICS) is a core theoretical course that develops the logical foundations essential to advanced areas in CS such as model checking, formal verification, and compiler design. The course introduces formal systems through propositional and predicate logic and gradually builds into computational logic topics like SAT solvers, resolution methods, temporal logic, and binary decision diagrams (BDDs).

While it covers a lot of theory, the course has practical relevance and trains students to reason clearly about systems and software. It is especially useful for those inclined towards CS research, formal methods, or systems-level thinking.

Primary Textbook: Huth & Ryan – Logic in Computer Science

Additional References: Baier & Katoen (Model Checking), Ben-Ari (Mathematical Logic)

Study Guide

LICS is not a course that rewards rote learning. Though most exams are open book, they're built around logical puzzles and proofs that demand pattern recognition and trained reasoning.

Initial weeks of class are especially crucial — this is where the foundations are laid: understanding truth tables, syntactic rules, natural deduction, and translating everyday statements into logic. These skills are the backbone for more advanced modules like SAT solving and model checking later in the semester.

Attending early lectures helps build the mental models needed to approach proofs naturally. Slides and notes are thorough, but real progress comes from solving. Go through every example from class, and revisit solved problems multiple times until you're fluent in moving between English and formal logic.

Assignments — typically a group project involving the implementation or modeling of logical inference — are a strong scoring component. Pick your group wisely and begin early, as these can take time but are also one of the most rewarding learning experiences in the course.

Tips to Succeed

- Practice deduction techniques like Modus Ponens, natural deduction, proof by contradiction, etc.
- Solve past year questions (PYQs) — they often follow similar logical forms.

- Don't underestimate predicate logic — translating quantified statements requires precision.

- Focus on building a mental library of logical patterns, not memorizing rules.

- Group discussions can really help solidify proof structures and eliminate blind spots.

Evaluative Components

The evaluation is spread across tests, a project, and major exams. While exams are open book, most questions can't be solved by simply looking up material — they test your ability to apply formal rules and perform multi-step deductions. Group projects typically involve implementing logical reasoning tools or systems, which reinforces theory through application.

You must manage your time well, especially in the comprehensive exam, which often tests soundness, completeness, and compactness using multi-part logic problems.

Grading & Course Average

Grading in LICS is generally lenient, provided you show consistent effort. Performing well in the group assignment and demonstrating understanding in your solutions (even if they're partially incomplete) is often rewarded. Class participation and seriousness in tutorials may help borderline cases.

Most students who stay consistent and solve problems actively through the term land in the A/A- range. Open-book does not mean easy — only those who have practiced enough will benefit from it.

Final Thoughts

LICS is one of the most intellectually stimulating CS core courses. It builds a way of thinking that sharpens clarity, precision, and structure in problem-solving. If you're considering systems, algorithms, verification, or research, this is a must-take. The early learning curve is steep, but once

you're into it, the logical elegance of the subject is deeply satisfying.

This course is about learning to think like a computer scientist — analytically, rigorously, and with complete logical clarity.

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Discrete Structures for Computer Science (CS F222) Course Content

DISCO is a mathematics-heavy CS core course that provides the essential logical and combinatorial tools used across computer science — from algorithms and data structures to databases, automata theory, and even AI. It covers propositional and predicate logic, proof techniques, combinatorics, recurrence relations, graph theory, trees, and basic algebraic structures.

The course is often described as “Maths for CS,” and while it can feel abstract, its relevance becomes increasingly clear as you progress through later courses. It builds not just mathematical fluency, but also the formal reasoning style essential in technical computing disciplines.

Primary Textbook: Mott, Kandel & Baker – Discrete Mathematics for Computer Scientists
Suggested Reference: K.H. Rosen – Discrete Mathematics and Its Applications (Highly recommended)
Study Guide

DISCO is best approached through regular problem-solving and not passive reading. The early weeks focus on proof methods — direct, contrapositive, contradiction, induction — which serve as the foundation for everything that follows. Once you're confident with these, topics like sets, functions, relations, and counting become easier to manage.

The second half of the course introduces graph theory, trees, and algebraic structures — areas where conceptual clarity matters more than memorizing formulas. Planar graphs, chromatic numbers, spanning trees, and BFS/DFS algorithms frequently appear in questions.

Classes tend to get optional once you're confident with the flow of content and are comfortable solving problems from Rosen or class notes. That said, the initial few lectures are worth attending to understand how concepts are introduced and connected.

Tips to Succeed<

- F Don't just read solved examples — solve them again yourself]
- F Induction and structural proofs need repeated practice]
- F Maintain a ready formula/reference sheet for combinatorics and graphs]
- F Group study helps, especially with graph theory and recurrence relations]
- F Practice PYQs early — they reveal common traps and favored proof types.

Evaluative Components

You'll face a midsem, three quizzes, and a comprehensive exam. Quizzes are closed book, short, and concept driven, and there's only one combined make-up quiz. Midsem and compre are proof-heavy and long. Time pressure is very real — efficiency in writing proofs matters just as much as correctness.

The compre especially is known to be lengthy and rigorous, testing multiple proof techniques and spanning all modules.

Assignments are rare, so most of your grade hinges on exam performance. There's no project or coding component.

Grading & Course Average

Grading is moderately strict. You're unlikely to get free marks, but the paper is always fair. Strong performance in the compre — where the weightage is highest — is crucial to push your grade upward. Students who solve enough problems regularly and approach topics conceptually tend to land in the A/B range.

Final Thoughts

DISCO is a classic CS foundational course. It trains you to think mathematically about problems and prepares you for logic-oriented courses that follow — like Algorithms, Automata, and even Machine Learning.

If you give it steady time and focus on practicing rather than memorizing, it becomes one of the most useful and intellectually satisfying courses in the curriculum.

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Object-Oriented Programming (CS F213)

Course Content

OOPS is a core programming course focused on building strong fundamentals in object-oriented programming using Java. It's split into two clear halves — the first is coding-heavy and focused on Java syntax, while the second dives into deeper OOP concepts like inheritance, polymorphism, abstract classes, interfaces, Java collections, generics, multithreading, and design patterns.

The course moves fast and demands consistent practice. It goes beyond syntax to focus on design — how objects interact, how software is structured, and how to make it scalable using patterns and principles like SOLID.

Primary Textbook: Herbert Schildt – Java: The Complete Reference

Secondary: Cay Horstmann – Object-Oriented Design & Patterns Study Guide

The early phase of the course covers basic Java — control flow, methods, classes, arrays, file I/O, exception handling, and overloading. It's essential to nail these topics early, because the second half assumes fluency in them.

After midsem, the course introduces core object-oriented ideas like inheritance hierarchies, method overriding, polymorphism, abstract classes vs interfaces, lambda expressions, collections (ArrayList, HashMap), and multithreading. Design patterns (like Singleton, Factory, Observer) are introduced late in the course and often come in lab tests or the compre.

The single lab test carries significant weight and tends to be the make-or-break component. It's conducted in a controlled, timed environment, often involving a mix of writing classes, implementing design patterns, debugging existing code, and tracing class relationships.

Tips to Succeed

- B Code regularly — use online IDEs like Replit or JDoodle to stay sharp
 - B Write clean, well-structured, and commented code
 - B Focus deeply on understanding inheritance, polymorphism, and interfaces — they are often confused
 - B Maintain handwritten notes of Java-specific syntax quirks and error patterns
 - B Treat the lab test like a practical final — do dry runs of design pattern implementations in Java.
- Evaluative Components

The course includes two short quizzes, a computer-based test (CBT), a midsem, a compre, and one major lab test. The lab test is extremely important and can significantly impact your final grade.

Design pattern understanding and object relationship clarity are frequently tested in exams and lab. Some weightage is also given to software design concepts like UML diagrams and class relationships in later modules.

Grading & Course Average

Grading is considered strict. A-cutoffs are often high — typically around average + 90 — so simply doing okay won't cut it. You need to score well in midsem and do very well in the lab test to land an A. Code correctness, structure, and object design are all evaluated seriously.

Final Thoughts

OOPS is one of the most practically relevant CS core courses. It sets the stage for building real-world software systems and directly influences your approach to problem solving, software architecture, and teamwork in future projects. It's not just about knowing Java — it's about thinking in objects.

A strong hold on this course gives you an edge in coding interviews, internships, and later project-based electives.

B.E. EEE/ENI/ECE **ULTIMATE**
Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Electronic Devices (EEE F214 / INSTR F214 / ECE F214)

Course Content

Electronic Devices (ED) is a foundational core course that bridges semiconductor physics and practical electronic circuit applications. It provides the physical and mathematical groundwork for understanding modern electronic devices such as diodes, BJTs (Bipolar Junction Transistors), and MOSFETs, all of which form the building blocks of analog and digital electronics.

The course begins by covering crystal lattices, semiconductor models, and carrier dynamics, then moves to junction theory, including the PN junction diode, metal-semiconductor contacts, and optoelectronic devices like photodiodes and LEDs. The later part of the course focuses on field-effect transistors (JFET, MESFET, HEMT, MOSFET) and BJTs, discussing their structure, IV characteristics, small-signal models, and secondary effects. The course concludes with optoelectronic devices and their practical roles in sensing and communication.

The lectures balance theoretical derivations with device behavior insights, and connect physical phenomena to circuit-level observations.

Concepts like carrier injection, recombination, diffusion, and electrostatic modeling are treated rigorously, making it a crucial course for students eyeing careers in microelectronics, VLSI, or semiconductor fabrication. The primary textbook is "Solid State Electronic Devices" by B.G. Streetman and Sanjay Banerjee, with Donald Neamen and Robert Pierret serving as excellent references for conceptual reinforcement.

Tips to Succeed

- W Never miss lectures. Prof. Navneet Gupta builds concepts sequentially and logically. His in-class explanations often go beyond slides and help solidify fundamentals.
- W Read from Neamen for building strong intuition, especially for early chapters on carriers and semiconductors.
- W Solve every tutorial seriously. Tutorial problems are directly reflected in quizzes and often reappear in modified form in exams.
- W Focus on derivations and physical interpretations, especially in the PN junction and MOSFET sections — many test questions revolve around these.
- W Practice visualizing band diagrams and energy transitions — these help with both qualitative and quantitative questions.
- W Memorize standard results like the diode equation, continuity equations, and small-signal models — you'll need them under time pressure.
- W Work through past year questions (PYQs) — many patterns repeat, especially in the midsem and comprez.
- W Group discussions for difficult topics (like MOSFET second-order effects or BJT small-signal models) can significantly boost understanding.

Evaluative Components

The evaluation is balanced across tutorial-based testing, midsem exams, and a comprehensive final. While the exams are closed book, they emphasize application of theory and quantitative reasoning.

W Tutorial Quizzes – 20%

Short-duration quizzes (15 min), often surprise in nature, held during tutorial hours. These typically test derivations, conceptual questions, and quick numericals.

W Mid-Semester Exam – 36.6%

90-minute test, generally moderate-to-hard in difficulty. Emphasizes fundamental understanding, with 1–2 direct questions and others requiring deep logical thinking.

W Comprehensive Exam – 43.3%

Includes both derivation-heavy and concept-application problems. Those who attend lectures and solve tutorials fare well. Emphasis is placed on BJTs, MOSFETs, and optoelectronic devices in the comprez.

W Total – 300 marks (Midsem + Quiz + Compre)

Grading & Course Average

Grading in ED is considered moderately lenient. It rewards consistency more than brilliance — students who attend class, complete tutorials, and revise steadily tend to do well. Quiz and midsem averages may seem low, but this is accounted for in relative grading.

W A @ Average + 65–66 mark

W A- @ Average + ~50 mark

W B @ Average + ~30 marks

You can safely expect a B+ or A- with regular effort and practice. Full marks in tutorials and decent scores in midsem/ comprez usually suffice.

Final Thoughts

Electronic Devices is a cornerstone course for anyone pursuing electronics, VLSI design, embedded systems, or semiconductor physics. Its content is rich and deeply interconnected, which makes consistent learning essential. The early weeks (covering carrier models and junctions) lay the bedrock — neglecting them makes later topics exponentially harder. If you're interested in how real-world electronic components actually function at the physical level, this course will be both challenging and highly rewarding. It may feel dense at times, but it provides the clarity needed to confidently approach advanced EEE/ECE electives.

B.E. EEE/ENI/ECE **ULTIMATE** Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Digital Design (CS F215 / ECE F215 / EEE F215 / INSTR F215) Course

Content

Digital Design is a core course in the undergraduate electronics and computer science curriculum that introduces students to the fundamental concepts of digital logic and circuit design. It blends theory with hands-on exposure, preparing students to understand, analyze, and implement combinational and sequential circuits — the building blocks of all digital systems.

The syllabus spans number systems, Boolean algebra, logic gates, and minimization techniques (K-Maps, Quine-McCluskey), followed by the design of combinational and sequential logic. It includes MSI circuits like multiplexers, decoders, and comparators, and moves on to flip-flops, registers, counters, and finite state machines (FSMs). Advanced topics include Register Transfer Level (RTL) design, Algorithmic State Machines (ASMs), and a basic introduction to HDLs such as Verilog or VHDL.

Practical sessions are integral — students gain experience with logic simulation tools and real hardware. The main textbook is M. Morris Mano & Ciletti, with the BITS EDD Lab Manual guiding practical work. This course is essential for students aiming at computer architecture, embedded systems, VLSI, or hardware development, and is foundational for advanced courses like Microprocessors and Computer Organization.

Tips to Succeed

T Master the basics: Boolean logic, gate-level design, and truth tables form the foundation. Complex circuits build directly on these.

T Understand the 'why': Know why specific flip-flops or counter types are chosen — don't design blindly.

T Don't overlook K-Maps: Seemingly simple, but easy to slip up on. Be comfortable with 4-variable and 5-variable forms.

T Focus on FSMs: Their design, simplification, and diagram-based questions appear often in exams.

T Practice basic HDL: Even limited syntax familiarity helps in lab and comprez.

T Lab preparation is key: Lab tests reward correct output and understanding — theory alone won't suffice T Use visual aids: Neso Academy, All About Electronics, and YouTube tutorials are great for conceptual reinforcement
T Try simulations at home: Tools like Logisim or Quartus are useful beyond lab hours T Work through PYQs: Question patterns, especially midsem and Part A of the compre, tend to repeat.

Evaluative Components

Evaluation balances theory and practice, ensuring both conceptual depth and real-world skills are tested T Tutorial Quizzes – 15%

Weekly quizzes, often simple if you stay consistent. Missing a few can hurt

T Lab Attendance – 10% 1 mark per lab; easy buffer if you're regular

T Lab Test – 35% Mid- or end-semester. Output-focused. Scoring above 25 is considered strong

T Mid-Semester Exam – 30% Closed book, 90 minutes. Usually tricky; expect logic puzzles and FSM applications T Comprehensive Exam – 40D

T Part A (40 marks) – closed book, concept-base

T Part B (80 marks) – open book, design-heavy: FSMs, ASMs, HDL coding

Total: 300 marks

Grading & Course Average

Grading depends heavily on lab and midsem performance, with room to recover in the compre T A @ Avg + 55–5<

T A- @ Avg + 45–4;

T B @ Avg + 15

Strong lab and Part B scores often make the difference. Midsem averages are generally low but manageable with practice.

Final Thoughts

Digital Design is a logic-centric, hands-on course that sharpens circuit thinking and system-level design skills. It's among the most practical courses for students heading into hardware, systems, or embedded fields. The lab component reflects real industry workflows, making it valuable beyond academics.

B.E. EEE/ENI/ECE **ULTIMATE** Sophomore Guide

Electrical Machines (EEE F211 / INSTR F211 / ECE F211)

Course Content

Electrical Machines is a fundamental core course that introduces students to the principles, characteristics, and applications of electromechanical energy conversion. It plays a crucial role in understanding how electrical power is generated, transformed, and utilized in real-world systems — from household appliances to industrial motors and power plants.

The course begins with a review of magnetic circuits and poly-phase systems, followed by an in-depth study of transformers (single- and three-phase), including construction, equivalent circuit, testing, efficiency, and voltage regulation. It then transitions to rotating machines, covering DC machines, three-phase and single phase induction motors, and synchronous machines. The syllabus also touches upon special machines like brushless DC motors and PM synchronous motors.

The course has a strong lab component, where students perform practical experiments to reinforce theoretical concepts — such as open/short circuit tests, Swinburne's test, load tests, and transformer connections. The prescribed textbook is I.J. Nagrath & D.P. Kothari, supported by the BITS lab manual. Electrical Machines is highly relevant for students interested in power systems, control, automation, and industrial applications. It forms the backbone for electives like Power Electronics and Electric Drives. Tips to Succeed

Z Solve textbook problems regularly: The theory is only half the battle — numerical problems from Nagrath & Kothari often appear directly in tests}

Z Don't skip tutorials: Many tutorial questions are reused or adapted in midsem/compre exams. They also reinforce lab concepts}

Z Understand the physical intuition: Knowing why certain machines behave the way they do helps greatly during derivations and conceptual questions}

Z Keep lab records updated: Many tutorial tests include questions related to the lab experiments} Z Pay attention during field visits (if arranged): Questions from these visits have appeared in tutorial tests and compre}

Z Review derivations for EMF, torque, and efficiency: These are frequently tested and essential to circuit analysis}

Z Use simplified circuit models for revision — drawing and analyzing equivalent circuits quickly is a valuable skill during exams.

Evaluative Components

The evaluation combines theory, tutorials, and lab performance, with a notable emphasis on hands-on understanding}

Q Tutorial Tests – 20%

Surprise quizzes during tutorials. Typically short and based on theory/lab concepts} Q Mid-Semester Exam – 25%

Open book, conceptual and numerical. Often considered tough — average scores tend to be low} Z Lab + Lab Test – 20%

Includes attendance, lab reports, viva, and a practical exam. Most students score well if they perform sincerely}

Q Comprehensive Exam – 35%

Contains both objective (fill-in-the-blanks, true/false) and subjective sections. Concepts from tutorials, labs, and textbook problems are tested.

Total: 300 marks

Grading & Course Average

Grading is based on relative performance, with tutorials and labs playing a key role in maintaining the average. The midsem is typically the lowest scoring component}

Z A @ Till Rank 2=

Z A- @ Till Rank 4;

Z B @ Rank 50 onward

Consistent lab work and strong tutorial performance can easily place you in the B+ to A- range, even if midsem scores are low.

Final Thoughts

Electrical Machines is a technically rich and practical course that connects theoretical electromagnetics with real-world applications. While the subject matter can feel dry or mechanical at times, its relevance to core electrical engineering is unmatched.

If you stay consistent with labs, practice textbook problems, and don't fall behind in tutorials, the course is manageable and rewarding — especially for those interested in industrial automation, energy systems, or electric vehicle technologies.

B.E. EEE/ENI/ECE **ULTIMATE** Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Electromagnetic Theory (EEE F212 / INSTR F212 / ECE F212)

Course Content

Electromagnetic Theory (EMT) is a mathematically intensive and concept-driven core course that forms the backbone of electrical, electronics, and communication engineering. It explores how electric and magnetic fields interact, propagate, and give rise to electromagnetic waves — the foundation for technologies like wireless communication, antennas, microwaves, and RF systems.

The course begins with transmission line theory, offering an accurate model for high-frequency circuits. This is followed by a mathematical review — vector calculus, coordinate systems, and integral theorems (Stokes', Gauss'). It then delves into electrostatics and magnetostatics, covering Gauss' and Ampère's laws, boundary conditions, and the behavior of dielectrics and magnetic materials.

The latter half focuses on Maxwell's equations, their time-varying and phasor forms, and how they describe EM wave propagation, reflection, and transmission. The course concludes with an introduction to radiating systems and antennas.

The prescribed textbook is David K. Cheng – Field and Wave Electromagnetics, with Hayt and Sadiku as popular references. Students also use LTspice for simulating transmission line problems, adding a practical component.

This course is essential for students interested in RF, microwaves, antennas, optics, and electromagnetic compatibility, and sets the stage for electives like Antenna Theory, RF Engineering, and Waveguides. Tips to Succeed

- s Attend every lecture: Prof. Praveen Kumar A.V. (PKAV) explains abstract concepts with strong physical intuition — and doesn't use slides|

- s Keep up with the math: A solid grasp of vector calculus (grad, div, curl, Laplacian) is crucial. Review first year material if needed|

- s Focus on physical understanding: Know what the equations mean — don't just memorize forms| s Use Cheng only as a reference: Start with class notes and expand with Cheng or Hayt when needed| s Practice tutorial problems: These often align closely with quiz and exam formats|

- s Manage your time during exams: Midsem and compre questions can be long — many students lose marks due to incomplete calculations|

- s Simulate transmission line problems: LTspice assignments are easy if you understand the basics. Avoid last-minute submissions|

- s Revise boundary conditions and wave propagation thoroughly — these are tested heavily in comprehensives.

Evaluative Components

The course evaluation is theory-heavy but includes simulation work for hands-on understanding. Exams are often lengthy but conceptually fair|

- \ Assignment (LTspice) – 5%

Simple if done with care. Focus is on correct setup and neat presentation|

- \ Surprise Quizzes – 16.66% Conducted during tutorial hours. Easy to score full marks if you revise tutorials regularly| \

Mid-Semester Exam – 30% Closed book, 90 minutes. Focuses on derivations, applications, and field problems.

Calculations can be time-consuming|

- \ Comprehensive Exam – 48.33% 3 hours, partly open book. Includes wave equations, Maxwell's laws, boundary conditions, and transmission line reflections.

Total: 200 marks

Grading & Course Average

Grading is moderately lenient, but the mathematically heavy content keeps averages low| s A @

Avg + 64–6

s A- @ Avg + 5G

s B @ Avg + (~35–40)

If you attend class, complete tutorials, and avoid calculation errors, an A- or B+ is easily within reach. Final Thoughts

Electromagnetic Theory is dense, abstract, and deeply foundational. It requires both mathematical skill and physical insight. While the learning curve can be steep, mastering EMT opens doors to high-impact fields like wireless tech, RF systems, and electromagnetics simulation.

If you're willing to engage with both the equations and the concepts behind them, this course becomes intellectually rewarding and highly applicable to real-world engineering challenges.

B.E. Mechanical

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Mechanics of Solids (ME F211) Course

Content

Mechanics of Solids is one of the most fundamental courses in Mechanical Engineering. It focuses on understanding how solid bodies respond to external forces and moments, and how internal stresses and strains develop within them. The course starts with the basics of statics and equilibrium, and progressively builds toward more advanced topics such as bending, torsion, deflection, and buckling of structural members.

The syllabus includes`

- _ Axial loading and deformation
- _ Trusses (including statically indeterminate ones)
- _ Shear force and bending moment diagram
- _ Mohr's Circle for stress and strain analysis
- _ Torsion of circular shafts (solid and hollow)
- _ Combined stresses and theories of failure
- _ Deflection in beams using integration and energy method
- _ Euler's buckling theory for columns

This course is not just essential for advanced mechanical design, but also forms the backbone for civil and manufacturing applications. The emphasis is on developing physical intuition through diagrams and derivations, and rigorous mathematical formulation through equilibrium equations and compatibility relations.

The primary textbook is Crandall, Dahl & Lardner, while Gere & Timoshenko is a preferred reference for additional problems and visual clarity.

Tips to Succeed

- _ Attend lectures diligently, especially Sharad Sir's and JSR Sir's — concepts are built in layers and their explanations make later topics easier
- _ Work through class slides and tutorial problems. A significant portion of midsem and comp questions are taken directly or are slight variations of these
- _ Understand the flow of forces visually. Drawing clear Free Body Diagrams (FBDs) is crucial
- _ Practice Mohr's Circle for both stress and strain until it becomes intuitive
- _ For deflections and torsion, ensure you memorize the standard formulas and understand where they come from
- _ Theories of failure and strain energy questions tend to be tricky but are highly scoring if prepared well in advance.

Evaluative Component

- _ Tutorial Tests – 20% (Best 4 out of 5; typically structured and solvable)
- _ Quiz – 10% (MCQ-based, conceptual and numerical)
- _ Midsem Exam – 30% (60 marks; proof-based + numericals)
- _ Comprehensive Exam – 40% (80 marks; includes OB section, slightly tougher)

Grading & Course Average

- _ A: Top 25 rank
- _ A-: Ranks 25–40
- _ B/B-: Based on class performance; Avg +15–20 marks is usually safe for a K
- _ Course Average: Generally low; understanding basics well can easily put you above average

Thoughts

This course is all about building strong fundamentals — visualizing forces, knowing when to apply equilibrium vs. compatibility, and writing stepwise, logically sound answers. Marks are awarded for method as much as for the final answer. Treat this as the gateway to all design and structure-based courses — and you'll appreciate its importance more in your higher semesters.

Materials Science and Engineering (ME F216) Course

Content

Materials Science and Engineering is a critical subject that connects the structure, properties, processing, and performance of engineering materials. It provides the foundation to understand why specific materials are chosen for particular engineering applications, how they respond under mechanical and thermal loads, and how they can be improved or modified.

The course covers a wide variety of materials including metals, polymers, ceramics, and composites. It delves into crystal structures, lattice imperfections, diffusion, phase diagrams, heat treatment, mechanical properties, and modern innovations like smart and nanomaterials.

Topics include]

- [Atomic bonding, crystal structures, and defect
- [Phase diagrams, solidification, and phase transformation
- [Mechanical behavior: hardness, toughness, creep, and fatigue
- [Heat treatment (TTT/CCT diagrams, quenching, annealing
- [Material characterization: XRD, SEM, TEM, NDo
- [Advanced topics: composites, polymers, ceramics, smart materials, and material selection principles

Textbook: Callister – Materials Science and Engineering

References: Raghavan, Dieter, Meyers – for deeper theory and real-world cases

Tips to Succeed

- [Make a summary sheet of definitions and phase diagrams — many quiz questions are based on recall
- [Focus on class-taught content — especially the slides and figures, as tutorial and exam questions often come directly from the~
- [Use the textbook for clarity on topics like crystallography, defects, and mechanical testing
- [Memorize key graphs like stress-strain curves, TTT/CCT diagrams, and phase diagram
- [For heat treatment and failure analysis, understand the physical meaning behind microstructural change
- [Lab component is important — prepare thoroughly for each experiment and viva

Evaluative Component

- [Midsem Exam: 25% (50 marks; closed book)
- [Tutorials (Best 2 out of 3): 10%
- [Laboratory Work + Reports: 15%
- [Lab Quiz/Viva: 10%
- [Comprehensive Exam: 40% (50% open book + 50% closed book)

Grading & Course Average

- [A: Around top 20–25%
- [A–: Top 25–35%
- [B/B–: At or slightly above average
- [Course Nature: High-average subject; easy to score well if consistent
- [Repeat PYQs: Many questions are repeated with minor changes in values

Final Thoughts

This is a conceptual and memory-intensive course that rewards visual understanding. The content forms the base for design, manufacturing, and failure analysis in mechanical systems. With solid notes, timely revision, and attention in labs, you can score high even without heavy cramming. A well-scoring and rewarding subject.

Applied Thermodynamics (ME F217) Course

Content

Applied Thermodynamics builds upon first-year Thermodynamics and shifts the focus toward real world engineering applications — particularly in power generation, refrigeration, and air conditioning systems. It trains students to analyze thermal cycles, evaluate system efficiencies, and estimate heating/cooling loads in buildings.

The course includes

- p Gas Power Cycles (Otto, Diesel, Brayton, Stirling, Ericsson, Dual, etc.)
- p Vapor Power Cycles (Ideal and actual Rankine cycles with reheating and regeneration)
- p Refrigeration and Heat Pump Systems (Vapor compression & absorption cycles)
- p Psychrometrics (Moist air properties, mixing, air conditioning processes, cooling/heating load estimation)
- p Thermodynamic Relations (Maxwell relations, Clapeyron equation, Joule-Thomson coefficient)

The focus is on evaluating thermal performance using P-v, T-s, and h-s diagrams. Students also perform real-world calculations involving boilers, turbines, condensers, cooling towers, compressors, and air-handling units.

Textbook: Çengel & Boles – Thermodynamics (8th Ed.)

References: P.K. Nag, R.C. Arora, C.P. Arora — useful for refrigeration, power plants, and psychrometry

Tips to Succeed

- p Attend lectures regularly — Prof. Suvanjan Bhattacharya explains numericals and cycle logic step by step
- p Make separate revision sheets for standard cycle efficiencies, psychrometric formulas, and refrigerant properties
- p Use T-s and Mollier diagrams during OB exams — mark key states and paths clearly
- p Solve PYQs and textbook examples for each type of cycle — they often repeat with small changes
- p Prepare lab reports honestly and learn experiment logic — lab viva and evaluation carry significant weight
- p Focus heavily on post-midsem content — psychrometry and refrigeration are lengthy but highly scoring

Evaluative Components

- p Midsem Exam: 25% (50 marks; closed book)
- p Quizzes (Best 3 out of 4): 15% (30 marks; closed book)
- p Lab Report + Quiz + Lab Examination: 25% (50 marks)
- p Comprehensive Exam: 35% (70 marks; open book)

Grading & Course Average

- p A: Around top 15–18 rank
- p A–: Top 18–30 rank
- p Average: Typically moderate due to length of paper; time management is key
- p Lab performance and consistency across all components strongly influence final grade

Final Thoughts

This is a numerical-heavy, application-oriented course that rewards systematic preparation and attentiveness in class. It's crucial for students interested in HVAC, energy systems, or thermal design. Focus on understanding the physical processes behind each cycle and not just memorizing formulas. If tackled with discipline, it becomes a high-scoring subject with immense practical value.

B.E. Mechanical

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Fluid Mechanics (ME F212) Course

Content

Fluid Mechanics is a core course that teaches how fluids (liquids and gases) behave under various conditions of flow and pressure. It is central to understanding and designing systems involving turbines, pumps, pipe networks, aircrafts, hydraulics, and more. The course builds both mathematical and physical intuition about how momentum, energy, and

mass are transferred in fluid systems.

Key topics include_

-] Mathematical Preliminaries: Scalars, vectors, tensors, vector calculus, Gauss' and Stokes' theorem] Fluid Statics: Pressure variation, forces on submerged surfaces, hydrostatics under acceleration] Reynolds Transport Theorem: Control volume analysis of mass, momentum, and energy conservation] Navier-Stokes Equations: Derivation, exact solutions, boundary condition
-] Dimensional Analysis: Buckingham Pi Theorem, model similarity
-] Inviscid Flow: Bernoulli's equation, stream functions, potential flow theory
-] Compressible Flow: Isentropic flows, shocks, Mach number
-] Internal & External Flows: Pipe flow, turbulence, boundary layer theory, drag and lift

Textbook: Fluid Mechanics by Frank M. White (8th Ed.)

References: Fox & McDonald, Potter & Wiggert — excellent for alternative explanations and practice problems

Tips to Succeed

-] Attend lectures regularly — Prof. Shyam Sundar explains numerical derivations that are not in book] Solve all class worksheets and tutorial problems — these are often reused in exam] Make a formula sheet for key laws — continuity, Bernoulli's, energy equation, Moody diagram values] Focus on pre-midsem content — it's well-structured and usually scoring
-] For compres, focus on compressible flow, boundary layers, and N-S derivation logii] Practice solving using dimensional analysis & Pi terms — a frequent source of 1–2 exam questions] Objective Quizzes are trick-based — revise theoretical definitions and curve trends thoroughly

Evaluative Component

-] Midsem Exam: 25% (50 marks; closed book)
-] Tutorial Tests (Best 2 out of 3): 20% (usually numerical and formula-based; announced in class)
-] Objective Quizzes (Best 1 out of 2): 10% (short, fast-paced, theory + trick questions)
-] Comprehensive Exam: 45% (Mixed: closed + open book; concept-heavy)

Grading & Course Average

-] A: Top 15–20 rank
-] A–: Top 20–30 rank
-] B/B–: Average typically low due to tough post-midsem content
-] Scoring Tips: Pre-midsem is easier and more scoring — secure marks early. Compre is lengthy and requires fast problem-solving.

Final Thoughts

Fluid Mechanics can feel overwhelming due to its mathematical depth, but it's incredibly rewarding if you understand the logic behind derivations and fluid behavior. The subject has strong real-world relevance — from aerospace to biomedical devices — and is often tested in higher electives and job interviews. Don't skip classes and focus on understanding why each formula works, not just how to apply it.

B.E. Chemical

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Chemical Process Calculations (CHE F211) Course

Content

Chemical Process Calculations is a foundational course in chemical engineering. It equips students with the core analytical tools required to perform material and energy balances for chemical processes. The course sets the stage for future process-oriented courses by introducing principles of stoichiometry, process flow diagrams, and thermodynamic properties.

The syllabus includes

- c Material balances (with and without chemical reactions)
- c Energy balances (for closed and open systems)
- c Recycle, purge, and bypass systems
- c Ideal gas and real gas behavior
- c Multiphase equilibrium and vapor-liquid systems

- c Humidity, saturation, and psychrometry
- c Thermophysical property estimation
- c Introduction to simultaneous material and energy balances

This course emphasizes clear thinking and structured problem solving. It lays the groundwork for process simulation and design, and connects closely with later courses like Process Technology, Reaction Engineering, and Separation Processes.

The primary textbook is Himmelblau & Riggs — Basic Principles and Calculations in Chemical Engineering (8th Edition), with Felder & Rousseau's Elementary Principles of Chemical Processes as a recommended reference.

Tips to Succeed

- c Be consistent with unit conversions and dimensional analysis — a large number of errors come from careless mistakes here
- c Memorize key definitions: mole fraction, mass fraction, specific gravity, etc
- c Understand the choice of basis in problems and be flexible in changing it when needed
- c Diagrammatic representation of flows (block diagrams, flowcharts) helps visualize complex multi-unit systems
- c Get comfortable with recycle, purge, and bypass scenarios — these are often asked in midsem and comprehensive
- c For energy balances, know the difference between open and closed systems, and practice enthalpy calculations
- c Humidity and psychrometric charts can be tricky — memorize the process steps and terminology.

Evaluative Component

- c Tutorial Tests – 20% (Best 4 out of 6; conceptual + numerical)
- c Surprise/Class Tests – 10% (Often short, best 5 considered)
- c Midsem Exam – 30% (Mix of theory + numerical)
- c Comprehensive Exam – 40% (Typically tougher, may include open-ended or multi-step problems)

Grading & Course Averages

- c A: Top ~25 rank
- c A–: Ranks ~25–40
- c B/B–: Based on overall distribution, 15–20 marks above average usually secure a X
- c Course Average: Generally low-to-moderate, especially in comprehensive years

Final Thoughts

CHE F211 is about thinking like a chemical engineer — understanding what's going into and coming out of a system. Build strong basics here, and you'll find future courses much more approachable. Diagrams, logical structure, and thorough unit handling are key to excelling.

B.E. Chemical

ULTIMATE

Sophomore Guide

Fluid Mechanics (CHE

F212) Course Content

Soumya Yadav (Msc Math + BE CS)

Fluid Mechanics is a core engineering science course that introduces students to the behavior of fluids in motion and at rest, and their interaction with boundaries and solids. As one of the pillars of transport phenomena in chemical engineering, this course equips students to analyze a wide variety of systems — from flowing pipelines and pumps to fluidized beds and aerodynamic drag. The course begins with the fundamentals of fluid properties and static systems, and gradually builds toward more dynamic and complex applications such as turbulent pipe flow, boundary layers, and compressible flow. This transition from conceptual foundations to applied engineering forms the backbone of several process design and equipment sizing decisions in industry.

The syllabus includes

- f Fluid properties: viscosity, surface tension, compressibility
- f Fluid statics: pressure variation, hydrostatic forces, buoyancy and stability
- f Conservation laws (mass and momentum): integral and differential form
- f Euler and Bernoulli's equations for ideal flow
- f Dimensional analysis and similitude (Buckingham Pi theorem)
- f Internal flows: laminar and turbulent flow through pipes, Moody chart usage
- f External flows: boundary layer theory, drag forces, flow over bodies
- f Flow through packed beds, fluidization, sedimentation
- f Pumps, valves, pipes, and flow metering
- f Introduction to compressible flow and flow measurement devices

The course is heavily mathematical in nature but emphasizes physical intuition. Developing a mental model of how fluids behave — when accelerated, squeezed, or sheared — is key to applying these principles to real-life systems.

The primary textbook is Fox & McDonald's Introduction to Fluid Mechanics, with McCabe, Smith & Harriott's Unit Operations serving as an excellent complementary reference.

Tips to Succeed

Visualize flows. Don't just solve equations — understand the physical significance of streamlines, velocity fields, pressure drops, and energy losses
For statics, practice applying hydrostatic laws in curved and inclined surfaces — diagrams help immensely here
In dynamics, master the control volume approach using Reynolds Transport Theorem. Being clear with system boundaries and assumptions is crucial
For Bernoulli's equation, understand where it applies (inviscid, incompressible, along a streamline) and when it doesn't
Dimensional analysis can be very scoring — learn the Buckingham Pi method step-by-step, and practice identifying repeating variables
Pipe flow problems are central: use the Moody chart confidently, and know how to compute major and minor losses (due to fittings, expansions, etc.)
When studying pumps and flow meters, pay attention to performance curves and efficiencies — these are frequently tested both conceptually and numerically
External flow and boundary layers are often confusing — focus on what boundary layer thickness and separation really mean
Practice derivations, visual interpretations, and numerical applications — the course balances all three types of assessments.

Evaluative Component

Tutorial Tests – 20%

(Best 6 out of 8; usually structured numerical or conceptual problems, often from textbook-type problems) Quizzes – 10%

(Best 3 out of 4; MCQs or short-answer questions, conducted during lectures)

Midsem Exam – 30%

(90-minute exam with both derivation-based and numerical questions)

Comprehensive Exam – 40%

(Includes open book + closed book sections; typically 3 hours and includes applied and slightly more challenging problems) Midsem grading is based on a combination of tutorial test and quiz performance, giving a chance to build a strong internal score even before the comprehensive exam.

Grading & Course Average

A Grade: Top 20–25 performers, often those who consistently solve tutorial sheets and understand derivations
A– Grade: Next 20–30 performers who perform steadily across all components

B/B– Grade: Usually those with average performance in numericals; scores of +15–20 over average are typically safe for a B
Course Average: Varies depending on the batch, but generally moderate, usually around 50–55%. The open book section of the comprehensive exam can lead to slightly tougher grading if you're underprepared, so rely on understanding, not just reference material.

Final Thoughts

Fluid Mechanics is not just about equations — it's about understanding the motion of invisible matter and quantifying its effects on systems and equipment. Whether you're sizing a reactor, designing a filtration unit, or analyzing pump operations, fluid mechanics will be right at the center.

Think of this course as your gateway to understanding "how things flow" in chemical engineering — whether gases in pipelines, liquids through reactors, or suspensions in columns. A solid grip on these principles pays dividends in later courses like Transport Phenomena, Process Design, and Reaction Engineering.

To succeed, combine a clear conceptual mindset with the discipline of problem-solving — and don't forget to enjoy the elegance behind how fluids move and behave!

B.E. Chemical

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Chemical Engineering Thermodynamics (CHE F213) Course Content

Chemical Engineering Thermodynamics is a core conceptual course that delves into the energetics and equilibrium of chemical processes. It introduces students to the fundamental laws that govern energy transformations, phase behavior, chemical equilibria, and the properties of substances.

While physics-based thermodynamics lays the foundation, CHE F213 adapts it for chemical process design and analysis. The course progresses from classical thermodynamic principles to real-world applications in chemical reactions, phase equilibrium, refrigeration, power cycles, and solution behavior — connecting energy, entropy, and equilibrium with real industrial implications.

The syllabus includes:

First Law of Thermodynamics: Energy balances for closed and open systems, heat capacity, enthalpy
Second Law & Entropy: Entropy generation, reversibility, Carnot efficiency, and the concept of availability (exergy)
Equations of State: Ideal gas law, virial and cubic equations (van der Waals, Redlich-Kwong, etc.)

Thermodynamic property relations: Maxwell's relations, residual and excess properties

Heat effects: Sensible heat, latent heat, standard enthalpy of formation and combustion

Power generation cycles: Steam engines, turbines, IC engines, jet propulsion

Refrigeration and liquefaction: Carnot and vapor compression cycles, refrigerant selection

Phase and Chemical Equilibria: VLE, fugacity, activity coefficients, Gibbs-Duhem relation

Solution Thermodynamics: Partial molar properties, mixing, excess Gibbs energy model

Chemical Reaction Equilibria: ΔG , equilibrium constants, Le Chatelier's principle in quantitative terms
The course is math-heavy, especially in partial derivatives, but what makes it powerful is the logical reasoning it develops — knowing whether a process is possible, favorable, or how it can be made more efficient.

The prescribed textbook is Smith, Van Ness, Abbott & Swihart – Introduction to Chemical Engineering Thermodynamics (8th Ed.), with Elliot & Lira and Koretsky as strong supplementary texts.

Tips to Succeed

Grasp the physical meaning behind laws and equations. Know what energy, enthalpy, and entropy really represent
The First Law is about bookkeeping (energy conservation), while the Second Law is about feasibility — get clear on this distinction

Be precise in derivations: Maxwell relations, property equations, and the chain rule from calculus appear frequently
Don't skip property diagrams (T-S, P-H, P-V) — they help visualize processes and cycle efficiencies

Memorize standard enthalpies and practice calculating ΔH , ΔS , and ΔG

using different temperature and pressure conditions
Cubic equations of state and fugacity can seem abstract — focus on their physical interpretation and calculation techniques

Understand how and when to use activity and fugacity coefficients for non-ideal systems

VLE and solution thermodynamics are frequent weak areas — practice Raoult's Law, modified Raoult's, and activity models (Margules, Van Laar, etc.)

In chemical equilibria, be comfortable with setting up equilibrium expressions and solving nonlinear systems (sometimes numerically)

Repeatedly solve tutorial problems, especially for thermodynamic cycles, entropy generation, and reaction equilibrium.

Evaluative Component

Tutorial Tests – 24%

(Best 4 of 5; usually involve problem-solving and conceptual short questions)

Midsem Exam – 30%

(Proof-based + numerical questions; a strong mix of derivations and thermodynamic applications)

Comprehensive Exam – 40%

(Covers all modules; tougher, often includes derivation-heavy questions and multi-part numerical problems)
Assignment – 6%

(Take-home; often conceptual or research-linked)

Make-up policy is strict; tutorial tests are not repeated, so consistent attendance is important.

Grading & Course Average

g A Grade: Reserved for the top 20–25 students who excel in both numericals and derivation

g A– Grade: For students just below the top, usually scoring consistently across tests and compr

g B/B– Grade: Often obtained with +15–20 marks above the average (strong internal score can boost chancesb g Course Average: Tends to be moderate-to-low, given the difficulty level — typically in the 45–55% range This is a course where conceptual clarity and derivational ability carry significant weight. Writing partially correct answers with correct logic often still earns partial credit.

Final Thoughts

CHE F213 is not just a course — it's the thermodynamic lens through which engineers see the world. Whether it's about predicting how a reaction proceeds, designing a refrigeration cycle, or determining feasibility of separation — thermodynamics is always involved.

Doing well in this course builds a deep foundation for future courses like Reaction Engineering, Process Simulation, Heat Transfer, and Separation Processes. More importantly, it trains you to think like an engineer — balancing ideal models with real-world limitations, and optimizing systems with efficiency in mind.

Don't fear the equations — embrace the story they tell. If you focus on visualizing systems, linking concepts logically, and practicing problems diligently, this course can be one of the most intellectually satisfying ones in your curriculum.

B.E. Chemical

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Engineering Chemistry (CHE F214) Course

Content

Engineering Chemistry is a wide-ranging, multidisciplinary course that introduces the essential chemical principles underpinning the engineering world. Unlike the heavily mathematical core chemical engineering courses, CHE F214 is more conceptual and application-driven, bridging the gap between theoretical chemistry and industrial processes.

It spans topics from organic and physical chemistry to materials science, analytical techniques, and environmental chemistry. The course lays the foundation for understanding materials, reactions, and processes commonly encountered in chemical engineering and related fields.

The syllabus includesZ

Y Organic Chemistry: Functional groups and their reactions (alcohols, aldehydes, acids, amines), important named reactions used in industry (Aldol condensation, Friedel-Crafts acylation, Hofmann rearrangement, etc.V

Y Physical Chemistry: Thermodynamic principles, enthalpy and entropy, electrochemistry, catalysis, adsorptiou Y Analytical Chemistry: Volumetric methods (acid-base, redox, complexometric titrations), instrumental methods (IR, UV-Vis spectroscopy, ChromatographyV

Y Engineering Materials: Polymers, refractories, cementing materials, lubricants, insulator

Y Metals & Alloys: Structure, properties, steel types, alloy composition

Y Fuels & Combustion: Types of fuels (solid, liquid, gaseous), calorific value, combustion reactions, flue gas analysi Y Corrosion:

Mechanisms, types, influencing factors, and prevention method

Y Water Chemistry: Hardness, impurities, water treatment techniques, boiler feed water specifications The course provides a holistic chemical perspective on everything from material selection and corrosion prevention to fuel analysis and water purification — making it highly relevant for practical engineering.

The primary textbook is Engineering Chemistry by Dr. Suba Ramesh et al. Supplementary references include Atkins (Physical Chemistry), Solomons & Fryhle (Organic), and Perry's Handbook for industry data.

Tips to Succeed

Y Take each module seriously — the course is broad but scoring. Most questions are directly from lecture content and standard referencesh

Y Memorization + logic is the strategy. Many sections rely on facts (like reactions, catalysts, or spectroscopy peaks), but logical understanding of why they happen helps retentionh

Y For organic chemistry, learn the mechanisms of reactions, not just the final products. Understand how functional groups behave in different environmentsh

Y Electrochemistry and thermodynamics sections require careful equation work. Get comfortable with Nernst equation, electrode potentials, and concepts like entropy and enthalpyh

Y Practice drawing and interpreting IR and UV spectra. Know key functional group peaks and common patternsh Y In fuel analysis and water treatment, numerical problems based on formulae (e.g., hardness, calorific value) are frequent — memorize standard units and equationsh

Y Corrosion and polymers are often theory-heavy but very scoring. Focus on classification, applications, and prevention methodsh Y For instrumental techniques, understand principle, construction, and application — this triad is often used in exam questions.

Evaluative Component\$

Y Surprise Quizzes – 25%

(Best 5 out of 6; mix of MCQs and short answers — focused on lecture contentV

Y Mid Semester Test – 25%

(90-minute closed book; includes both theory and applied questionsV

Y Term Project – 15%

(Team or individual assignment on industrial chemistry topics — e.g., fuel types, water treatment, corrosion controlV Y Comprehensive Exam – 35%

(Final 3-hour exam — mainly closed book; covers all modules with short/medium-length questions) The quiz weightage is high, so being regular with lectures and revising each topic weekly gives a big advantage.

Grading & Course Average

Y A Grade: Students who combine consistent quiz scores with strong compre and midsem performanc Y A– Grade:

Earned through good internal performance and moderate-to-strong compr

Y B/B– Grade: Typically achievable if you maintain average marks across all evaluation

Y Course Average: Generally above average, often between 55–65%, making this a scoring course with focused effort Because of its objective and semi-theoretical nature, even students from non-chemistry backgrounds often perform well with basic preparation and attention to lecture content.

Final Thoughts

CHE F214 is about real-world chemistry — the reactions, materials, and processes that underpin engineering as we know it. From the water in boilers to the steel in bridges, from fuel combustion to corrosion control — chemistry is everywhere, and this course teaches you to understand it deeply.

Unlike more calculation-intensive courses, this one rewards conceptual clarity, memory, and application. It's an excellent opportunity to gain exposure to interdisciplinary topics that will appear in courses like Environmental Engineering, Materials Science, and Process Safety.

Use this course to broaden your perspective and appreciate the “chemistry” behind the engineering you'll do in labs, plants, and design projects.

B.E. Manufacturing

ULTIMATE Soumya Yadav (Msc Math + BE CS) Sophomore Guide

Mechanics of Solids (MF F211) Course

Content

Mechanics of Solids is one of the most fundamental courses in Manufacturing Engineering. It focuses on understanding how solid bodies respond to external forces and moments, and how internal stresses and strains develop within them. The course starts with the basics of statics and equilibrium, and progressively builds toward more advanced topics such as bending, torsion, deflection, and buckling of structural members.

The syllabus includes

- Axial loading and deformation
- Trusses (including statically indeterminate ones)
- Shear force and bending moment diagram^{1/2}
- Mohr's Circle for stress and strain analysis^{1/2}
- Torsion of circular shafts (solid and hollow)
- Combined stresses and theories of failure[±]
- Deflection in beams using integration and energy method^{1/2}
- Euler's buckling theory for columns

This course is not just essential for advanced mechanical design, but also forms the backbone for civil and manufacturing applications. The emphasis is on developing physical intuition through diagrams and derivations, and rigorous mathematical formulation through equilibrium equations and compatibility relations.

The primary textbook is Crandall, Dahl & Lardner, while Gere & Timoshenko is a preferred reference for additional problems and visual clarity.

Tips to Succeed

- Attend lectures diligently, especially Sharad Sir's and JSR Sir's — concepts are built in layers and their explanations make later topics easier
- Work through class slides and tutorial problems. A significant portion of midsem and comp questions are taken directly or are slight variations of these
- Understand the flow of forces visually. Drawing clear Free Body Diagrams (FBDs) is crucial Practice
- Mohr's Circle for both stress and strain until it becomes intuitive
- For deflections and torsion, ensure you memorize the standard formulas and understand where they come from
- Theories of failure and strain energy questions tend to be tricky but are highly scoring if prepared well in advance.

Evaluative Component[±]

- Tutorial Tests – 20% (Best 4 out of 5; typically structured and solvable)
- Quiz – 10% (MCQ-based, conceptual and numerical)
- Midsem Exam – 30% (60 marks; proof-based + numericals)
- Comprehensive Exam – 40% (80 marks; includes OB section, slightly tougher)

Grading & Course Average[±]

- A: Top 25 rank^{1/2}
- A–: Ranks 25–40
- B/B–: Based on class performance; Avg +15–20 marks is usually safe for a k
- Course Average: Generally low; understanding basics well can easily put you above average Final

Thoughts

This course is all about building strong fundamentals — visualizing forces, knowing when to apply equilibrium vs. compatibility, and writing stepwise, logically sound answers. Marks are awarded for method as much as for the final answer. Treat this as the gateway to all design and structure-based courses — and you'll appreciate its importance more in your higher semesters.

B.E. Manufacturing **ULTIMATE**
Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Transport Phenomena in Manufacturing (MF F218) Course Content

Transport Phenomena in Manufacturing is a core engineering course that plays a vital role in developing foundational understanding for various manufacturing processes. Taught by Prof. Anesh A. M., the course introduces students to the principles of fluid mechanics, heat transfer, and mass transfer, focusing on their application in real-world manufacturing systems. It is split into two logical segments^V

The pre-midsem portion emphasizes fluid statics, kinematics, dynamics, and introduces the powerful Reynolds Transport Theorem, which serves as the backbone for analyzing transport processes

The post-midsem portion covers the three modes of heat transfer — conduction, convection, and radiation, followed by special topics such as heat exchangers, multi-mode heat transfer, simultaneous heat and mass transfer, and phase change.

The course begins with a conceptual introduction, building up from fundamental definitions and principles, and gradually shifts into mathematical modeling and real-world problem solving. Students learn how to derive governing equations for mass, momentum, and energy transport, apply appropriate boundary conditions, and explore both analytical solutions and empirical correlations for standard engineering problems.

Prof. Aneesh emphasizes understanding over rote memorization. Marks are awarded more for a clear problem-solving approach than just getting the final answer. Hence, partial credit is common and generous if the student shows correct application of concepts and logical steps.

While the lecture slides and textbooks (Welty, Bird, Incropera, etc.) provide good theoretical support, some content is exclusively covered in class, often discussed verbally with small problem-solving tricks or conceptual clarifications — making lecture attendance essential. The slides alone are not sufficient for the midsem or compre.

The labs are focused on applying the theory to practical systems. These include experiments on heat exchangers, convective cooling, conduction, and radiation setups. Labs also have quizzes and reports, and there's no buffer unless stated during orientation. Vivas can feel intense, but their weightage is small — confidence in concepts will suffice. Surprise quizzes and tutorial assignments are scattered throughout the semester. These can be scoring opportunities if you're consistent with lectures and practice. Assignments are often derivation-based or involve applying concepts from class to new problem settings.

Tips to Succeed~

Understand the derivations deeply, especially those related to fluid statics, Reynolds theorem, and 1D/2D heat conduction. Practice solving standard problems on pipe flow, external convection, and radiation — these are frequently tested. Class attendance is crucial — a lot of "what to focus on" is shared verbally and may not appear in official slides. Stay up to date with labs, and review the related theory before entering the lab session.

Solve previous year questions and assignments — the pattern remains largely consistent year to year. During open-book exams, ensure your notes are well-organized and curated — flipping through 300 pages of textbook will waste time.

Evaluative Components

Tutorial Quizzes & Assignments – 20E

Surprise format, announced in class

Lab Practicals & Quizzes – 20E

Includes report writing, 2 quizzes, and possibly a viva

Mid-Sem Exam – 25E

Open-book; problem-solving focused

Comprehensive Exam – 35E

Format may be open or closed-book depending on IC instruction

Typically includes derivations + applied numericals

Grading & Course Average

Grading is generous if you're consistent; focus is on approach and understanding

6–7 students typically receive an A, followed by ~5 A– grade

Marks are fairly distributed, and mistakes in calculations don't hurt much if your logic is sound

Final Thoughts

Transport Phenomena in Manufacturing is a well-designed, intellectually rich course that ties engineering theory with manufacturing relevance. It's demanding in the sense that it requires you to understand and apply, but it's forgiving to those who stay consistent. With good lecture discipline, focused lab engagement, and a strong grip on derivations and boundary conditions, this course can become both scoring and genuinely insightful. For anyone pursuing core mechanical or manufacturing roles in the future, it lays down a highly relevant conceptual foundation.

B.E. Manufacturing

ULTIMATE

Sophomore Guide

Machine Drawing (MF

F217) Course Content

Soumya Yadav (Msc Math + BE CS)

Machine Drawing, is a lab-centric, design-focused course that immerses students in the practical aspects of engineering drawing and 3D modeling. It is one of the most hands-on courses in the 2-1 Mechanical curriculum and is taught by Dr. Amit R. Singh, who brings an industry-aligned approach to the learning experience. The course is structured to help students not only understand mechanical drawing conventions but also develop real CAD skills that are directly applicable in internships, industry projects, and future coursework.

The course begins with manual and digital sketching, teaching students how to create fully constrained 2D profiles. This foundational phase helps in establishing control over geometries and introduces the idea of design intent — a crucial concept in parametric modeling. As the course progresses, students are introduced to solid modeling techniques, including operations such as extrusion, revolution, lofting, boolean operations, mirror reflections, patterns, threading, and filleting.

Once confident with part modeling, students move into 2D drawing generation, where they learn to create orthographic projections, sectional views, and accurate dimensioning following standard tolerancing conventions. These drawings are expected to align with industry norms and are created directly from their 3D models.

The later part of the course focuses on assembly modeling, motion studies, interference checks, and exploded views. Students learn to import components, define constraints, simulate motion, and analyze the physical interaction between parts. This culminates in sheet metal design and graphic rendering, where they work on enhancing the visual appeal of their models using textures, materials, and lighting effects.

The CAD tool used is typically Fusion 360, a cloud-based platform with powerful modeling and rendering capabilities. While the course uses a specific tool, the skills are transferable to other industry-standard software like SolidWorks, CATIA, or Creo. Importantly, the course is entirely practical — there are no traditional theory lectures or a written final exam. The entire assessment is based on performance in labs and a capstone project, making it highly performance-driven. Students are expected to complete their modeling tasks during lab hours and submit well-documented outputs that include both models and drawings.

Lab sessions are evaluative, and missing even one can be difficult to recover from, as no make-ups are allowed. TAs and the instructor are available for in-lab consultation, and actively asking for help is encouraged — the course rewards effort and willingness to learn more than just raw modeling skill.

Tips to Succeed!

- Never miss lab sessions — every session contributes to your grade, and there are no second chances
- Be proactive with learning Fusion 360 shortcuts, workspace tools, and navigation — modeling speed is crucial
- Focus on constraints and accuracy in sketches — sloppy modeling or bad design logic can cost marks
- Keep referring to the prescribed textbook (Jensen et al.) for standard drawing practices
- Take the final project seriously — it carries 30% of your grade and is a great opportunity to showcase creativity
- Use rendering tools to make your project visually strong — small details like lighting and material textures are noticed
- Maintain a clean documentation folder for all your model files, images, and drawings — organized submission matters.

Evaluative Component

- Mid-Semester Exam – 20%
 - Conducted during lab hours, CAD-based modeling and drawing task (open-book)
- Lab Evaluations – 50%
 - Weekly submissions during lab sessions; graded on accuracy, completeness, and adherence to design rule
- Final Project – 30%
 - Build a complete mechanical assembly in Fusion 360; must be original and not copied from existing templates

Topics (As per handout):

- Sketching – 2D fully constrained geometry
- Solid Modeling – 3D part creation using constructive operations and transformation
- 2D Drawing – Orthographic and sectional views, dimensioning, tolerance
- Assembly – Joints, constraints, interference/motion analysis
- Sheet Metal Design – Flattened templates and folded form
- Rendering – Visual enhancement using textures, colors, materials

Grading & Course Average~

- Grading is generous, provided you're consistent and submit all components
- Most students who finish their labs and do a proper final project secure an A or A+
- Marks are awarded for effort, accuracy, and attention to design standards, not just perfection

Final Thoughts

MF F217 is a highly rewarding and skill-oriented course. It removes the burden of theoretical exams and instead focuses on helping you build a real design portfolio. With consistent effort and a proactive attitude, you can come out of this course not just with a good grade, but also industry-ready CAD skills. Whether you're aiming for design internships, manufacturing roles, or mechanical product development, this course lays a strong foundation. Think of it not as just a CDC, but as an investment in your professional toolkit.

B.E. Manufacturing

ULTIMATE Soumya Yadav (Msc Math + BE CS) Sophomore Guide

Materials Science and Engineering (MF F216) Course

Content

Materials Science and Engineering is a critical subject that connects the structure, properties, processing, and performance of engineering materials. It provides the foundation to understand why specific materials are chosen for particular engineering applications, how they respond under mechanical and thermal loads, and how they can be improved or modified.

The course covers a wide variety of materials including metals, polymers, ceramics, and composites. It delves into crystal structures, lattice imperfections, diffusion, phase diagrams, heat treatment, mechanical properties, and modern innovations like smart and nanomaterials.

Topics include~

- Atomic bonding, crystal structures, and defects
- Phase diagrams, solidification, and phase transformation
- Mechanical behavior: hardness, toughness, creep, and fatigue
- Heat treatment (TTT/CCT diagrams, quenching, annealing)
- Material characterization: XRD, SEM, TEM, EDX
- Advanced topics: composites, polymers, ceramics, smart materials, and material selection principles

Textbook: Callister – Materials Science and Engineering

References: Raghavan, Dieter, Meyers – for deeper theory and real-world cases

Tips to Succeed!

- Make a summary sheet of definitions and phase diagrams — many quiz questions are based on recall
- Focus on class-taught content — especially the slides and figures, as tutorial and exam questions often come directly from the
- Use the textbook for clarity on topics like crystallography, defects, and mechanical testing
- Memorize key

graphs like stress-strain curves, TTT/CCT diagrams, and phase diagramÀ | For heat treatment and failure analysis, understand the physical meaning behind microstructural changeÀ
| Lab component is important — prepare thoroughly for each experiment and viva

Evaluative Component«

- | Midsem Exam: 25% (50 marks; closed bookx
- | Tutorials (Best 2 out of 3): 10e
- | Laboratory Work + Reports: 15e
- | Lab Quiz/Viva: 10e
- | Comprehensive Exam: 40% (50% open book + 50% closed book)

Grading & Course Averagi

- | A: Around top 20–2
- | A–: Top 25–3
- | B/B–: At or slightly above averag
- | Course Nature: High-average subject; easy to score well if consisten¼
- | Repeat PYQs: Many questions are repeated with minor changes in values

Final Thoughts

This is a conceptual and memory-intensive course that rewards visual understanding. The content forms the base for design, manufacturing, and failure analysis in mechanical systems. With solid notes, timely revision, and attention in labs, you can score high even without heavy cramming. A well-scoring and rewarding subject.

B.E. Civil

ULTIMATE

Sophomore Guide

Mechanics of Solids (CE

F211) Course Content

Soumya Yadav (Msc Math + BE CS)

Mechanics of Solids (MoS) is a fundamental course offered jointly to Civil, Mechanical, and Manufacturing students. It introduces the basics of how materials respond to external forces — focusing on stress, strain, deformation, bending, torsion, and axial loads. In the Civil section, the course is taught by Prof. Dipendu Bhunia (HOD, Civil) and is considered to be moderately conceptual with relatively lenient grading.

The curriculum includesV

- U Simple stress and strain, elastic constant
- U Shear force and bending moment diagram
- U Bending and shear stresses in beam
- U Torsion in circular shaft
- U Deflection of beam
- U Principal stresses and Mohr's circl
- U Columns and buckling

The content aligns with what Mechanical and Manu students study, so inter-section resources and cross-learning are helpful.

Tips to Succeej

- U All tutorial tests are open book and announced in advance, making it easy to prepare with printed slides and formula sheetsb
- U If Civil lectures feel unclear, attend Mechanical/Manu section lectures — the content is the same but may be explained differentlyb
- U YouTube videos and standard textbooks (like R.K. Bansal or Gere & Timoshenko) are good supplements for practice and understandingb
- U Leverage inter-department differences — Civil and Mech tests occur on different days, so sharing preparation resources is commonb

U Focus on drawing accurate shear force and bending moment diagrams — they're frequent in tests
U Memorize standard formulas but also understand why and when they apply
U Always show steps and diagrams clearly in your answers — presentation matters in partial marking.

Grading & Evaluation

U Grading is rank-based, making it highly relative. Here's the approximate breakdown

C U A:
Ranks 1–2

U A–: Ranks 26–30

U B: Ranks 31–40

U B–: Ranks 46–50

Because the grading is generous and based on performance rank, it is possible to score well with focused effort.

Final Thoughts

Mechanics of Solids in Civil is conceptual but not overwhelming. With its open-book tests, flexible lecture options, and shared syllabus, this is one of the more manageable courses. Build a good visual and logical understanding, and you'll be able to do well without heavy memorization.

B.E. Civil

ULTIMATE

Sophomore Guide

Surveying (CE

F213) Course Content

Soumya Yadav (Msc Math + BE CS)

Surveying is one of the most scoring and practical-oriented courses in the Civil Engineering curriculum. It focuses on the measurement and mapping of physical features on Earth using both traditional and modern tools. Students learn how to collect, analyze, and represent spatial data, which is essential for construction, land assessment, and geotechnical projects.

The syllabus covers

- T Chain surveying, compass surveying
- T Leveling and contouring
- T Theodolite usage and angle measurement
- T Total station surveying and GPS basic
- T Traverse calculations and plotting
- T Error handling and adjustment method
- T Field data collection and plotting on maps

What makes Surveying unique is the lab component, which happens outdoors on lawns and fields, simulating real fieldwork conditions.

Tips to Succeed

- T Theoretical classes are optional — attendance isn't crucial as most questions come directly from slides, which are sufficient for tests
- T The lab is crucial: Be punctual and take caps, water bottles, and sunglasses. The sessions are long, physically tiring, and done under the sun
- T All tests except midsem, lab comp, and comp are open book. Take printed slides to every test; questions are often copied directly
- T Tests are lengthy — even completing 3 out of 6 questions is considered good. Don't panic if you can't finish everything
- T You can often study during the test itself due to the open-book format — this works if you know where to find things quickly
- T There are make-up labs, but no buffers — missing a lab without a valid reason can be costly
- T Focus on understanding instruments and interpreting readings. Practical familiarity helps during the lab comp.

Grading & Evaluation

T Based purely on rank and relative performance, with an easier curve due to the open book format

T A: Top 1N

T A-: Ranks 16–2;

T B: Ranks 21–4N

T B-: Ranks 46–50

Because many students don't take the tests seriously or skip labs, simply attending regularly and putting basic effort can easily put you in the top grading bracket.

Final Thoughts

Surveying is a blessing GPA-wise. With predictable questions, open-book flexibility, and a practical lab that simulates real civil work, it offers one of the best opportunities to score an A with manageable effort. Prepare your slides, show up for labs, and stay hydrated — and you're already halfway to a top grade.

B.E. Civil

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Fluid Mechanics (CE F231)

Course Content

Fluid Mechanics is a challenging and core course in the Civil curriculum that deals with the behavior of fluids at rest and in motion. It is central to understanding systems involving water, air, and other fluids — like pipelines, dams, bridges, and even underground water flows.

The course includes a

- ` Properties of fluids (density, viscosity, surface tension]
- ` Hydrostatics: pressure variation, manometers, buoyancy
- ` Kinematics: streamlines, flow classification
- ` Dynamics: Bernoulli's theorem, impulse-momentum principle
- ` Laminar vs turbulent flow in pipe
- ` Losses in pipelines, major and minor losses, Moody chart
- ` Flow through open channel
- ` Introduction to dimensional analysis and similarity

While conceptually rich, students who lack strong JEE Physics fundamentals may find the course difficult. It demands attention, practice, and familiarity with equations and flow principles.

Tips to Succeed

- ` Hydrostatics is crucial. DO NOT MISS those classes — they form the base of 25–30% of the questions
- ` Buy the recommended textbook and practice numericals regularly. Fluid Mechanics is not just theory — you must work through problems
- ` Attend classes regularly. Even though the prof doesn't cover everything in depth, he points out the exact topics you need to prepare for exams
- ` Maintain a good rapport with the professor. He often reveals the exam questions or important hints in chamber consultations. Go pick up your corrected test papers — it builds goodwill
- ` If you're struggling, get help from friends in Mechanical or Chemical sections — fluid mechanics is common across branches
- ` Focus on formulae and when to use them — especially for Bernoulli's, energy equations, and flow measurement
- ` In tests, aim to solve at least one or two questions well. Due to low averages, even 5–6 marks can get you above average (av+)
- ` Papers are sometimes undersubscribed — many students skip tests. Just showing up and attempting sincerely can significantly improve your grade.

Grading & Evaluation

` One of the toughest courses to score in, due to difficulty level and test unpredictability

` A-: 21–25

` B: 26–40

` B-: 46–50

Average marks can drop to zero, so relative performance matters more than absolute score. Even 5–6 marks in

a test can move you into an av+ zone if the class average is poor.

Final Thoughts

Fluid Mechanics in Civil is not an easy A — it demands effort. However, the difficulty also works in your favor: if you attend class, make a little effort to understand flow principles, and show up for tests, you're likely to outperform most.

Start early with Hydrostatics, maintain a working relationship with the prof, and practice numericals consistently. These small habits can help you ace one of the most feared Civil courses.

B.E. Civil

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Civil Engineering Materials (CE F230) Course

Content

Civil Engineering Materials is a conceptual and visual course that introduces students to the physical, chemical, and mechanical properties of materials used in construction. It emphasizes recognition, classification, and understanding of behavior of materials under various conditions — making it an essential subject for future structural and geotechnical engineers.

The course typically covers

- N Classification and properties of construction material}
- N Bricks, concrete, cement, aggregates, timber}
- N Steel and non-ferrous metal}
- N Bituminous materials and pavement}
- N Admixtures, composites, and glass}
- N Material testing methods}
- N Modern materials and sustainable alternatives

Unlike calculation-heavy courses, this one is theory- and image-driven. Questions often involve visual recognition of materials, processes, or instruments — making attention to detail essential.

Tips to Succeed

- N Attend all classes. This is one of the few courses where anything said in class can appear in the tests. Professors frequently ask spontaneous or minor points mentioned verbally.
- N Every Friday = Test Day. Unless explicitly cancelled by the professor, expect a test every Friday. Be consistent with weekly prep — skipping one can hurt internal scores badly.
- N Don't rely on a textbook. Instead, focus on
 - N Slides shared by the professor
 - N Photos and videos embedded in the slides
 - N In-class notes
- N Questions are often of the type: "Recognize this material from the image" or "State the property/application shown in the figure."
- N Solve past year questions — they repeat often. Focus especially on the format and structure of the questions.
- N Maintain a collection of important images and diagrams for quick revision — visual memory plays a huge role here.
- N Treat it like a GK + observation-based course: instead of formulas, you'll be asked to identify, compare, or recall subtle details.

Grading & Evaluation

N No fixed cutoff system; grading is based on gap grading, i.e., performance distribution.

N Due to frequent tests and minor fluctuations, grades can swing quickly — consistency matters more than isolated performance.

In general, students who attend regularly and pay attention to slides/images tend to be in the top band.

Final Thoughts

Civil Engineering Materials is one of the most manageable and high-return subjects in the Civil curriculum — if taken seriously from the start. Weekly tests + image-based questions + professor's in-class hints make it ideal for students who are attentive, observant, and consistent. You don't need a book. You need your eyes, your memory, and your slides.

Msc. Economics

Economic Environment of Business (ECON F214)

Course Content

EEB is a highly relevant and moderately scoring economics course that explores how macroeconomic conditions and government policies shape the broader business landscape. It focuses on understanding economic indicators and applying policy frameworks to analyze real-world business situations, making it a crucial course for students interested in policy, consulting, or business strategy.

The course is conducted by Prof. Balakrushna Padhi, whose lectures are known to go beyond the shared slides. His teaching style involves discussion-based delivery, where key concepts, exam tricks, and case examples are explained orally and not documented in the official materials — making class attendance essential for success.

The syllabus generally includes

- X National Income Accounting (GDP, GNP, NDP, etc.)
- X Inflation, Interest Rates, Unemployment
- X Business Cycles and Economic Growth Model
- X Fiscal Policy and Budget
- X Monetary Policy and RBI's role
- X Exchange Rates and Balance of Payment
- X Trade Policy, Globalization, and WTB
- X Economic reforms and business environment in India

The course includes both qualitative (framework-based) and quantitative (data-driven, numerical) questions, with a strong emphasis on current economic events and their implications for businesses.

Tips to Succeed

- X Be Regular in Class: Padhi Sir intentionally excludes important content from slides. These missing pieces — often discussed verbally — appear in quizzes, midsems, and the comp. Missing classes can cost you critical marks!
- X Make Detailed Class Notes: Since slides are often restricted during open-book exams, neatly written and indexed handwritten notes become your most valuable resource!
- X Treat the Group Project Seriously: You will likely have to prepare a group assignment and class presentation. RSs may ask critical follow-up questions. Ensure everyone in the group understands the content to defend it well!
- X Do All Past Year Questions (PYQs): The exam pattern does not change drastically. Case-based or application-oriented numericals often repeat with slight changes in context or data!
- X Practice Data Interpretation: Some questions involve interpreting charts or economic data points. Get familiar with real-life GDP/inflation data to make your answers more relevant!
- X Review Short Notes Before Each Component: Exams can draw from any topic. A quick, focused revision using concise, topic-wise summaries is the most efficient last-minute strategy.

Grading & Evaluation

Grading in EEB is relatively competitive, especially since many students put in strong effort given the practical nature of the course!

- X A: Top 10%
- X A-: Next 10–15%
- X B: Next 20–25%
- X B-: Around the average score

The average grade is typically B-, meaning you need to stand out through your notes, consistency, and PYQ accuracy to make it to the A-range.

The evaluation is often composed of

- X Quizzes / Class Test
- X Midsem (mostly conceptual + numerical)
- X Group Assignment / Presentation
- X Comprehensive Exam (case-based + numerical)

Some years, section-specific RSs evaluate subjectively, so keep your formatting and language clean and structured. Final

Thoughts

EEB is a balanced course — not too heavy on numericals, not too dry on theory — and very aligned with real-world thinking. If you stay consistent with lectures, take organized notes, and prepare your group project well, this course can significantly boost your GPA and strengthen your economic reasoning.

It also helps build a foundation for electives like Public Finance, Global Economy, or Strategy & Policy. So don't treat this as just another HSS course — it's one where engagement pays off directly.

Msc. Economics

Principles of Economics (ECON F211) Course

Content

PoE is the foundational economics course at BITS that introduces students to both microeconomics and macroeconomics in a structured, concept-driven manner. Whether you're pursuing an economics minor or simply want to build your economic literacy, PoE is the base upon which all other economics courses are built.

The course is typically taught by Prof. Debi Prasad, and while class attendance is not mandatory, tutorials are crucial, as that's where real question-solving happens. The structure of the course leans more toward numerical and graphical problem-solving than theoretical writing, especially in midsem and compre.

The syllabus generally includesP

] MicroeconomicsP

-] Demand, supply, and market equilibriu
-] Elasticit|
-] Utility, consumer behavio
-] Production and cos
-] Market structures: perfect competition, monopoly, oligopol|

] MacroeconomicsP

-] GDP and national income accountinc
-] Inflation, unemployemen
-] Fiscal and monetary polic|
-] IS-LM and AD-AS model
-] Economic growth and international trade

The course integrates textbook theory, graphs, numerical problems, and real-world economic context, offering a well-rounded economics primer.

Tips to Succeed

-] Tutorials = Priority. Most students score well because they attend tutorials and solve the PYQs and tutorial sheets thoroughlyh
-] Class attendance is optional, especially under Debi Prasad. But if you're weak in a topic, it's best to watch recorded lectures fromP
 -] Rajan Pandey – crisp numerical clarit|
 -] Geetilaxmi / Rahul Arora – great for conceptual understandinc
-] Make structured notes with key graphs and solved examples — these can be used in open-book exams (especially for quizzes and tutorials)h
-] PYQs are gold. A large number of questions in tests are repeats or slight variations. If you cover the past 5–6 years' papers well, you're likely to walk into the exam with a strong edgeh
-] Focus on graphs and intuition: whether it's a supply-demand curve or cost structure in monopoly, drawing accurate graphs with labels gets you marks even if the rest is shakyh
-] Conceptual clarity trumps mugging. You don't need to memorize definitions — you need to understand economic behavior and apply basic math to it.

Grading & Evaluation

Grading is relatively structured and based on cumulative performanceP

-] A: Top 10;
-] A–: Next 10;
-] B: Next 15;
-] B–: Roughly at average

The average grade is usually B–, but the course is considered a GPA booster because a well-prepared student can comfortably score above average through tutorials and PYQs.

Evaluation components may include9

-] Quizzes (often open bookC
-] Tutorial performanc
-] Midsem (mostly numericals, graphsC
-] Comprehensive Exam (broad coverage — both macro and micro)

Time management during exams is important — some papers are lengthy, especially in the compre.

Final Thoughts

PoE is a high-yield and high-logic course. If you enjoy thinking about how markets behave, how businesses price their products, or how policies affect inflation and unemployment — this course will make sense and feel rewarding. To score wellP

-] Attend tutorials religious|
-] Practice PYQs thorough|
-] Master graphical representatiot
-] Watch recorded lectures for clarity

With these habits, PoE becomes one of the easiest A's in the economics stream and a stepping stone toward electives like Microeconomics, Development Economics, and Industrial Organization.

Msc. Economics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Fundamentals of Finance and Accounting (ECON F212) Course

Content

FundaFin is a structured, numerical-heavy course that introduces students to the world of financial statements, accounting principles, and corporate finance. It's often one of the first real exposure points for BITS students to understand how companies track money, measure performance, and assess financial health. The course is typically split into two phasesX

Pre-Midsem: Taught by Krishna M., focusing on the accounting half — balance sheets, journal entries, ledger posting, and income statementsw

Post-Midsem: Handled by Aprajita Pandey, covering financial analysis and decision-making — cash flow statements, ratio analysis, time value of money, and capital budgeting.

Topics includeV

Basic accounting principles and GAA`

Double-entry bookkeeping and journal entrie

Preparation of financial statements: balance sheet, profit & loss account, cash flo}

Depreciation methods and inventory valuatioj

Ratio analysis (liquidity, solvency, profitabilityR

Time value of money (NPV, IRRR

Capital budgeting and investment decisions

This course is extremely useful for those interested in consulting, finance, or entrepreneurship, but it does require consistent numerical practice.

Tips to Succeed

Attend all classes, especially the early accounting ones by Krishna M. He doesn't share slides later, and the lectures are the only source of explanation for many fundamental conceptsw

Aprajita Pandey's classes post-midsem involve intensive problem-solving. Try to mirror her solving methods and adopt her format in your test solutions — clarity and structure matterw

Do as many PYQs as possible. This course has repeatable and formula-based patterns, especially in quizzes and midsemw

Learn the logic behind each formula (e.g., why depreciation is subtracted, or how working capital impacts cash flows). Don't just memorizew

Be very careful with formats of financial statements — even small formatting errors can cost marksw Time management is key: the exams are lengthy, and papers can be overwhelming if you haven't practiced under time pressurew

Make a formula and format cheat sheet before the exam — many components are mechanical and scoring if you're well-prepared.

Grading & Evaluation

Grading is tougher than PoE or EEB, as many students underestimate the effort needed for numericals and struggle with lengthy papersX

A: Top 15>

A–: Next 10>

B: Based on gap, often competitiv

Average Grade: Typically a C

Don't take this course lightly — many students think it's theory-heavy and get surprised by how number-dense and calculation-heavy it becomes, especially in the compre.

Evaluation may includeX

Tutorial Tests / Quizze

Midsem (mostly journal entries and statement preparationR

Post-midsem tests on finance portioj

Comprehensive Exam (combination of accounting + finance questions)

Final Thoughts

FundaFin is a course where discipline beats intelligence. You don't need prior finance experience — just regular practice, careful attention in class, and a strong grip on past year questions.

If you want to enter fields like consulting, VC, investment banking, CA, or startup finance, this course gives you the first taste of what real-world financial thinking looks like. Get the basics right, and this course could be a game changer — both GPA-wise and skill-wise.

Msc. Economics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Mathematical and Statistical Methods (ECON F213) Course

Content

MSM is a concept-heavy, mathematical economics course that integrates tools from calculus, statistics, optimization, and game theory to develop decision-making frameworks. It is foundational for anyone looking to pursue higher-level economics, analytics, or data science electives.

The course is taught by Prof. Rahul Arora, who is known for his clear teaching style, especially in the Game Theory module, where his explanations are particularly appreciated.

The syllabus includes[

© Regression Analysis: Simple and multiple linear regression, R^2 , hypothesis testin]

© Probability & Statistics: Random variables, distributions, expectation, varianc

© Optimization: Unconstrained and constrained maxima/minima, Lagrangian metho

© Game Theory: Dominant strategies, Nash equilibrium, mixed strategie

© Calculus Applications: Partial derivatives, concavity/convexity, utility maximization

MSM combines mathematical rigor with economic intuition — you're not just solving equations but interpreting them in real-world economic contexts.

Tips to Succeed~

© Attend Game Theory lectures without fail. Rahul Arora explains it intuitively, and this is often the most heavily weighted and scoring portion of the compre

© Start weekly practice early, especially for calculus-based questions. MSM rewards consistency more than last minute prep

© Use SP Gupta as a reference for statistics and regression. It has clear examples and is aligned with the course flow

- © For Game Theory, refer to any introductory game theory text or online lecture (e.g., MIT OpenCourseWare) to build confidence
- © Understand the intuition behind concepts like Nash equilibrium, regression coefficients, or optimization conditions — don't just memorize methods
- © In exams, show proper steps: define variables, state assumptions, and interpret your answers. This often leads to partial marks even if you don't arrive at the final result
- © Use past year questions to identify important question types — MSM exams tend to follow a pattern with numerical-heavy sections.

Grading & Evaluation

Grading for MSM follows a similar structure to other economics courses like PoEJ

- © A: Top 10%
- © A-: Next 10%
- © B: Next 15%
- © B-: Around the course average

The average tends to hover around B-, but the difficulty level is slightly higher due to the mathematical nature of the course.

The evaluation structure typically includes

- © Tutorial Sheets / Test
- © Quizzes (often numerical)
- © Midsem Exam (calculus/statistics-heavy)
- © Comprehensive Exam (includes Game Theory, optimization, regression)

MSM demands balance between theory and application — both understanding and solving ability are tested. Final Thoughts

MSM is one of those courses where conceptual clarity + regular effort = guaranteed performance. While it might feel technical at times, the course actually teaches you how to think like an economist using math. It's a stepping stone for electives in data analysis, econometrics, behavioral economics, industrial organization, and research-based fields. Mastering MSM gives you an edge in both your GPA and your analytical abilities. To do well

- © Follow Rahul Arora's lectures closely
- © Start problem-solving early
- © Use books and PYQs together
- © Focus on both math and meaning

Msc. Mathematics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Discrete Mathematics (MATH F213) Course

Content

Discrete Mathematics (DM) is a core CDC for B4 students, overlapping significantly with the CS CDC Discrete Structures. It's widely regarded as extremely useful for placements, coding interviews, and theoretical CS. For B4A7 students, this course also replaces the need to take MATH F243 (Graphs and Networks) in the next semester, offering flexibility in future CDC planning.

The course is taught by Prof. Krishnendra Shekhawat, known for a fast-paced delivery, especially post midsem. The content builds logically, starting from foundational logic to more involved combinatorics and recurrence relations.

Key topics include

-] Propositional and predicate logic
-] Proof techniques (induction, contradiction, contrapositive)
-] Sets, functions, relation
-] Combinatorics and the Pigeonhole Principle
-] Recurrence relations and their solution
-] Basics of number theory
-] Introductory graph theory (limited — full coverage in 2-2 CDC)

Tips to Succeed

-] Attend the initial classes diligently — this is when the "rigorous solving structure" is introduced. Even if an answer feels intuitive, the professor expects a step-by-step logical structure in your solution
-] Slides are enough post-midsem, but ask for clarification if specific steps aren't mentioned clearly — mathematical rigour is a grading criterion
-] Use the textbook for practice, especially for recurrence relations and the Pigeonhole Principle — these are the sources of the trickiest problems
-] Midsem covers Module 1 to 4, typically up to inhomogeneous recurrences. Post-midsem moves faster, but builds on the logic from earlier modules
-] Practice solving questions in multiple ways — the instructor allows multiple methods as long as they are logically sound and complete
-] Most tests are announced MCQ-style with negative marking — avoid guesswork unless you're sure

Shekhawat's pace is fast, so review the slides and examples weekly to avoid falling behind.

Grading & Evaluation

Grading is relatively generous, especially compared to other B4 CDCs, due to

] Large enrollment (~90–100 students), including CS students doing DS minor or taking DM as an OPem] A mix of students with strong CP/math backgrounds (phoda seniors tend to grab many A's) Grading trend (approximate)

] A: 140–1509

] 10 people usually get an

] Better than other B4 2-1 CDCs, especially for those with CS/DCP prep

Final Thoughts

DM is highly logical and rewarding, especially for those with an analytical mindset or programming inclination. It's not just a B4 requirement — it's a foundational course for algorithmic thinking, and forms the core of data structures, logic, and advanced mathematics in CS and Pure Math tracks. Treat it as more than a math course — it's a problem-solving discipline. With early class attendance, stepwise logical reasoning, and a grip on recurrence/combinatorics, this course is very much scorable and high-utility.

Msc. Mathematics

ULTIMATE

Sophomore Guide

Optimization (MATH

F212) Course Content

Soumya Yadav (Msc Math + BE CS)

Optimization is widely considered the easiest B4 2-1 CDC in terms of content, pace, and grading — but only if approached with discipline. It introduces students to Linear Programming Problems (LPPs) and techniques to solve them efficiently using methods like Simplex, Transportation, and Assignment problems.

This is not the trivial +2 level LPP — while the core ideas may seem familiar, the execution involves longer calculations, iterations, and conceptual understanding of duality, degeneracy, and sensitivity.

The course is generally taught by either

e Prof. Rakhee (slightly stricter evaluator)

e Prof. Gaurav Dwivedi (GD) — considered a blessing due to his lucid teaching and very lenient grading Core topics covered

e Formulation of LPP

e Graphical and algebraic (Simplex) method

e Artificial variables (Big-M and Two-phase)

e Duality theorem

e Sensitivity analysis

e Transportation and Assignment problem

e Integer and Goal programming (overview level)

Tips to Succeed

e Pre-midsem: The course is very slow-paced in the beginning and focuses heavily on the Simplex method and its variations. Master this technique early — it's the backbone of the rest of the course

e Post-midsem: Application-heavy, but more manageable if your fundamentals are strong

e Do not take the course too lightly — many students lose marks in silly arithmetic mistakes, especially in Simplex iterations

e Go to class even if it's optional — most question types and tricks are practiced in lectures

e If you miss classes or need a reference, use:

Chandrashekhhar Sir's YouTube Playlist (Highly Recommended)

e The Taha textbook is good for understanding the theory, but solving class problems regularly is usually sufficient for tests

e Tut tests are very easy — do not lose marks due to lack of practice. They are often numerical and scoring.

Grading & Evaluation

Because Optimization is also a 3-1 EEE CDC, enrollment is large (~200+), and grading is generally the best among all B4

2-1 CDCsw

- e A Grade: Given at ~140 marks onward
- e ~20+ students typically secure an g
- e GD's section = leniently evaluate±
- e Rakhee's section is slightly stricter but still fair

Many students score full marks in Midsem or Compre. A well-prepared student can secure an A comfortably.

Final Thoughts

Optimization is the most scoring CDC for B4 in 2-1. If you're looking for one course to anchor your SG, this is it. But don't get complacent — lengthy calculations, time mismanagement, or silly mistakes can ruin your paper. Consistent practice, especially of Simplex and assignment-type problems, can make this a sure shot A.

Use this course to build a strong GPA cushion and confidence boost during an otherwise heavy semester.

Msc. Mathematics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Elementary Real Analysis (MATH F214) Course

Content

Elementary Real Analysis (ERA) is your first formal introduction to pure mathematics and proof writing. It forms the foundational backbone of the B4 (Math) curriculum and is revisited — either directly or indirectly — in future Pure Math CDCs like Measure Theory, Advanced Calculus, and Topology. ERA explores what makes mathematics rigorous, starting from basic number systems and building up to the logic of calculus. Instead of solving for answers, you're expected to prove that answers exist — with full logical clarity.

The course is taught byM

- V Prof. Sangeeta Yadav, o
- V Prof. Gaurav Dwivedi (GD) — known for clearer lectures and student-friendlier teaching.

Topics typically includeF

- V Logic, quantifiers, and basic proof technique
- V Sequences and their convergenc
- V Bounds, supremum/infimu
- V Countability and cardinalit
- V Real number construction and completenes
- V Series convergenc
- V Introduction to limits and continuit
- V Basic Riemann Integration (post-midsem)

The journey is slow at first, but becomes significantly more abstract and proof-heavy after midsem. Tips to

Succeei

- V Attend lectures — especially early in the course. They teach how to structure proofs, not just what to provec
- V Proofs require rigour. Even if your answer is “right,” you'll lose marks without full justification or if you skip stepsc
- V Expect subjective and unannounced tut tests, especially around sequences, limits, and convergence/ divergencec
- V Learn how to write standard proofs (especially for limits, cardinality, sequence convergence). A few standard long proofs are frequently repeated in Midsem/Comprec
- V Post-midsem, Integration becomes the most important module, both in weightage and difficultyc
- V Use lecture notes + slides as primary resources. The book (Kenneth Ross) is useful for deeper understanding, but not mandatory for scoringc
- V If you get stuck in the beginning, refer to online lectures or simpler proof guides. Once you understand the logic flow, the subject becomes more approachable.

Grading & Evaluation

ERA is known for rigid grading due to the difficulty and abstractness of the subject. In a typical B4 batchM V A Grade:

- Usually only 5–6 students get i
- V A–: 4–5 students mor
- V Average Grade: Often C, can fall to C– if class performance is weak

Grading is performance-dependent, i.e., the better the class performs, the more generous the curve. Unfortunately, most batches underperform due to the shift from numerical math to abstract logic.

Final Thoughts

ERA is not about what you prove — it's about how you prove it. If you're someone who enjoys building things from scratch and understanding the logic of math, this course will be intellectually satisfying. But if you don't stay consistent, it can feel alienating. SoM

- V Attend class
- V Master the standard proof pattern
- V Don't be afraid to ask for clarification
- V Revise weekly — don't cram this course

If approached right, ERA builds your mathematical maturity — and that's a skill you'll carry across all of B4.

Msc. Mathematics

ULTIMATE

Sophomore Guide

Algebra (MATH

F215) Course Content

Soumya Yadav (Msc Math + BE CS)

Algebra is your second pure math CDC in Sem 2-1, and it dives deep into abstract algebraic structures. Think of this course as a generalization of Number Theory — where the focus shifts from numbers to sets with operations, like groups and rings. It's often perceived as more abstract and difficult than ERA, and for many B4 students, the most challenging course of the semester.

The course is taught by

- U Prof. Divyanshu Sharma
- U Prof. Jitendra Kumar

Both are known for rigorous instruction and set high standards for presentation, clarity, and proof-writing. The syllabus includes

- U Group Theory (Pre-midsem)
 - U Binary operations, subgroups, cyclic groups, group homomorphism
 - U Cosets, Lagrange's theorem, normal subgroups, quotient group
- U Ring Theory (Post-midsem)
 - U Rings, ideals, ring homomorphisms, polynomial ring
 - U Integral domains, fields, and related structures

This course introduces you to formal mathematical structures, and often involves proof-based derivations, theoretical properties, and conceptual problem solving.

Tips to Succeed

- U Attend all lectures. Unlike Optimization or DM, this course is nearly impossible to grasp through slides alone. The lecture walkthroughs help you understand how abstract algebra "builds up"
- U Get comfortable with abstract definitions — "what is a group?" will go far beyond "a set with an operation"
- U The textbooks Gallian and Herstein are your best friends
 - U Gallian is great for intuitive understanding and examples
 - U Herstein aligns more with the IC's style and is deeper/more concise
- U Solve problems regularly. This is NOT a course where you can mug proofs — you'll have to derive, reason, and explain clearly
- U Mathematical rigour is critical. Avoid hasty conclusions or assuming things that aren't formally stated
- U Tutorials usually involve numerical or fill-in-the-blank MCQs — often with negative marking
- U Midsem and Compre are tough, and usually go beyond what's been spoon-fed in class. Make sure you've done
 - U Examples from Gallian/Herstein
 - U Class illustrations

Grading & Evaluation

This course is known for the harshest grading among the 2-1 B4 CDCs. In recent years

- U Only 1 A was awarded at 1304
- U A- was rare or absent
- U Most students landed in the C or C- range

Evaluation is performance-based: if your batch as a whole performs well, grading improves. But historically, many struggle due to the abstractness and proof requirements.

Final Thoughts

Algebra is not plug-and-play math. It's theoretical, elegant, but difficult. If you stay consistent, attend all classes, and respect the logic of abstraction, you will do better than most.

Think of it as a mental gym for your proof-writing and abstraction skills. Those who succeed in this course build solid ground for electives like Galois Theory, Linear Algebra (Advanced), and Algebraic Structures. To survive and thrive

- U Practice proof-writing weekly
- U Never assume — always justify
- U Use Gallian to understand, Herstein to refine
- U Don't fall behind — this course snowballs quickly

Msc. Physics

ULTIMATE Sophomore Guide

Classical Mechanics (PHY F211) Course

Content

Classical Mechanics (PHY F211) is a core physics course that offers a rigorous and formal introduction to advanced mechanics, far beyond what is taught in high school or JEE preparation. It builds the theoretical framework of Newtonian physics and then transitions into more elegant and generalized formulations such as the Lagrangian and Hamiltonian formalisms.

The course explores how to describe physical systems mathematically using generalized coordinates, constraints, energy principles, and conservation laws. It plays a foundational role in preparing students for upper-division physics courses like Quantum Mechanics, Statistical Mechanics, and Electrodynamics.

Typical topics covered

- I Review of Newton's Laws and Systems of Particle
- I Conservation of Momentum, Angular Momentum, and Energy
- I Central Force Motion (e.g., planetary orbits, effective potentials)
- I Simple Harmonic Motion and Oscillation
- I Calculus of Variations and Lagrangian Mechanics
- I Generalized Coordinates and Constraint
- I Hamiltonian Mechanics (phase space, canonical equations)
- I Rigid Body Motion (moments of inertia, Euler's equations)

You'll deal with fewer "plug-and-play" problems and more proof-based derivations, system modeling, and interpretation — especially in the second half of the course.

Tips to Succeed

- I Go to class regularly in the beginning — especially in the first 2–3 weeks. The professor often discusses proofs and problems in class that appear almost directly in Tutorial Sheets and Midsems, just slightly reworded
- I Focus on understanding derivations step-by-step, especially in Lagrangian and Hamiltonian mechanics. Even if a result seems intuitive, writing it without justification costs marks
- I Don't ignore the Assignments — there are usually two, one before Midsem and one before Compre. These assignments often contain the exact or rephrased questions that appear in the respective exams
- I If you are more comfortable with self-study, watch PhysicsDarshan on YouTube for the Pre-Midsem portion. It is reliable and concise
- I Post-Midsem topics are harder (especially the Lagrangian formulation, generalized coordinates, and rigid body dynamics). Here, reading the textbook (if you're patient) will significantly deepen your conceptual clarity
- I Class notes and slides should be revised regularly. Keep a separate notebook or document for proofs — many are reused, and memorizing their logical structure helps during time-pressured exams
- I Practice writing full-length derivations instead of short solutions. Presentation and structure of logic matter greatly in this course.

Grading & Evaluation

Evaluative components are largely predictable and revolve around your ability to replicate and apply classroom derivations and assignments

- I Tut Tests: Often feature proofs or derivations done in class. Usually short but scoring — consistency is key
- I Midsem: Usually well-aligned with Assignment 1 + class notes. Some professors directly lift questions from assignments or modify them slightly
- I Compre: Slightly tougher. Focus on Post-Midsem theory + Assignment 2. Practice PYQs, as certain derivation patterns or system setups often repeat.

A student who attends class, completes assignments properly, and does moderate PYQ revision can easily score above average.

Final Thoughts

Classical Mechanics (PHY F211) may seem intimidating due to its mathematical formalism and depth, but it's also one of the most intellectually rewarding courses in the curriculum. You'll learn not just "how" but "why" physical systems behave the way they do.

It's especially critical for students planning to pursue physics as a major, or anyone headed toward research, theoretical modeling, or high-level problem solving.

To make the most of this course

- I Don't ignore derivations. Practice writing them from scratch
- I Use class time efficiently — record important insights, questions, or examples
- I Take the assignments seriously — they are direct indicators of exam questions
- I Don't cram — mechanics needs time to mature in your head. Start early.

With consistent effort and a clear understanding of logical steps, this course can go from a fear-factor to a GPA-booster.

Msc. Physics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Electromagnetic Theory 1 (PHY F212) Course

Content

EMT-1 is one of the core second-year physics courses focused on building a rigorous, mathematical understanding of electrostatics and magnetostatics, forming the foundation for full Maxwell's equations in EMT-2. This course takes the ideas of electric and magnetic fields — introduced in high school — and explores them using vector calculus, differential equations, and formal derivations.

Despite its intimidating name, EMT-1 is considered relatively easy compared to Classical Mechanics. The course follows the textbook closely, and if the instructor is Prof. Debashis Bandyopadhyay, it's even more manageable — thanks to his well-organized lectures and detailed notes.

Typical syllabus includes

- Coulomb's Law and Electric Field
- Gauss's Law and Application
- Electric Potential and Poisson's Equation
- Capacitance and Dielectric
- Current Density and Continuity Equation
- Biot-Savart Law and Ampère's Law
- Magnetic Field of Current
- Vector Analysis (div, grad, curl) — often reviewed at the beginning

Both Pre-Midsem and Post-Midsem portions are well-aligned with the textbook, and many questions come straight from it.

Tips to Succeed

If Prof. Debashis is teaching

His class notes are gold — he provides elaborate lecture notes + book solutions. For many students, this is all you need to score well

You can treat the course as self-study friendly, but his notes will save time and effort

Going to class is optional, but if you prefer structure and don't want to rely too heavily on self-discipline, attending lectures helps keep you on track

Focus early on vector calculus fundamentals. Many problems use div/grad/curl identities. If you're weak here, spend time reviewing

During prep

Practice from the textbook if you're not attending class

Focus on problems suggested by the professor

Do PYQs — many are conceptually or numerically repeated

If you miss class, borrow notes from a friend or download complete solutions for the standard text (e.g., Griffiths or Sadiku) — these align well with the course content

Don't ignore theory questions — many tests ask for derivations or proof-based explanations (e.g., derivation of field due to a uniformly charged ring).

Grading & Evaluation

Tut Tests: Usually straightforward. Based on textbook questions and concepts. Very scorable if you've gone through the book or class notes

Midsem: Mostly drawn from class examples and book exercises. Do Assignment 1 thoroughly if given

Compre: Similar to midsem in format, just a bit lengthier. Includes PYQs + textbook-based derivations. Students who go through class notes and do suggested textbook problems tend to score very well. This is a relatively high-yield course if approached properly.

Final Thoughts

EMT-1 is not difficult if you approach it with a plan. The mathematical machinery (vector calculus) may feel slightly heavy at first, but the actual content is highly formulaic and derivation-based.

With organized notes, regular review of solved textbook problems, and basic PYQ practice, EMT-1 becomes a CG booster course. It's also essential for core physics courses later on (like EMT-2, Quantum Mechanics, and Electrodynamics).

If taught by Prof. Debashis, it's a gifted opportunity to score well without overwhelming pressure.

Msc. Physics

ULTIMATE Sophomore Guide

Optics (PHY F213)

Course Content

Soumya Yadav (Msc Math + BE CS)

Optics is widely regarded as the most relaxed and high-scoring course in the 2-1 Physics curriculum. It offers a balanced mix of conceptual understanding, diagram-based reasoning, and formula-based derivations, all centered around the behavior and nature of light.

Unlike Classical Mechanics or EMT-1, which are heavy on formalism or vector calculus, Optics is more intuitive. Most topics can be visualized, and if you've enjoyed wave optics or ray optics in school, this course will feel like an extension — but with more derivations, diagrams, and slightly deeper theory.

The syllabus typically includes

- Geometrical Optics: Reflection, Refraction, Lens systems, Optical instrument

W Interference: Young's Double Slit, Thin Film Interference, Newton's Rings, Michelson Interferometer W

Diffraction: Single slit, double slit, multiple slit, diffraction grating

W Polarization: Types of polarization, polarizers, Malus' law, scattering

W Fiber Optics / Optical Devices (sometimes taught as an extra or bonus module)

The focus is more on deriving and applying known formulas and interpreting results rather than building deep new theoretical frameworks — making it the most “approachable” CDC in this semester.

Tips to Succeed

W Attend classes especially if you don't enjoy reading the textbook — most students find the book dry or too formal, while lectures simplify the topics

W That said, PYQs + class notes = 90% of your preparation. This combo is more than sufficient if reviewed properly W Do not ignore derivations. Many questions — especially in Tut Tests and Midsem — are based on direct derivations or diagram-based explanations covered in class

W Focus on understanding

W Conditions for constructive/destructive interference

W Fringe patterns in Young's Double Slit

W Working of polarizers and interferometer

W Practice drawing neat and accurate diagrams — optics is a visually-driven subject, and presentation matters in subjective questions

W If your professor gives a list of suggested book problems, don't ignore them — many midsem questions are often lifted from there

W For the Comprehensive Exam

W If it's open book, be warned: such papers are often interpretation-based and indirect

W Memorizing where formulas are located won't help unless you understand when and why to use them W If it's closed book, expect slightly more direct questions (but sometimes trickier numerically) W Focus your effort on Tut Tests and Midsem — they're straightforward and easier to score in. Even if you find Compre tricky, a strong internal score will anchor your grade.

Grading & Evaluation

The evaluation is highly predictable and student-friendly if you play it right

W Tutorial Tests: Based on class questions, derivations, and standard proofs. Usually short and scoring — don't miss them

W Midsem: Do all PYQs, class notes, and textbook derivations. Stay focused on presentation and logical explanation W Compre: Can be slightly tougher or longer (especially if open book), but still manageable with prior preparation. Even an average-performing student who prepares systematically from PYQs and assignments can land a B+ or A-. The course's structure rewards consistency over brilliance.

Final Thoughts

Optics (PHY F213) is your best opportunity for a GPA boost among the three theory-based physics courses in 2-1. It combines manageable content, predictable questions, and light mental load — provided you are disciplined. You don't need to grind through long problem sets or memorize dozens of theorems. What you do need is

W Solid grip on 6–7 important derivations

W Neat diagrams and good presentation

W Regular PYQ practice

Treat it as the CGP anchor of your semester, especially if you're juggling tough CDCs like Classical Mechanics or ERA. With minimal stress, smart strategy, and basic effort, Optics becomes one of the easiest A's you can secure in your second year.

Msc. Physics

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

EMO Lab (Electromagnetism & Optics Lab – PHY F214) Course

Content

EMO Lab is the most relaxed and scoring component of the 2-1 Physics curriculum. It complements the theory courses (Optics and EMT-1) by allowing students to perform hands-on experiments related to light, electric/magnetic fields, and basic instrumentation.

The course is less about deep theoretical understanding and more about

a Following lab protocols carefully

a Answering viva questions with confidence

a Presenting neat, timely, and accurate reports/plots

You'll typically perform 1 experiment per week, often in groups. Each experiment includes

a Pre-lab readings (from lab manual)

a Experimental procedure and observation

a Viva (oral test) during the lab

a Post-lab report submission

Experiments include

a Verification of Malus' Law (Polarization)

a Measurement of Wavelength using Newton's Ring

a Resonance in LCR Circuit

a Study of Laser Diffraction

a Calibration of Galvanometer

a Determination of Refractive Index using Prism method

a EM-based measurements like mutual inductance, etc.

The labs are standardized, and performance depends more on presentation and instructor perception than difficult problem solving.

Tips to Succeed

- a Choose your instructor wisely — some instructors (like Srijata Dey) are known to be stricter in viva grading. Peers usually share which instructor is lenient or helpful
- a Go through the lab manual at least 1 hour before the lab session. Understand the objective, circuit/diagram, and expected results
- a Viva questions are mostly factual or procedural
 - a What is the principle of this experiment~
 - a What formula are you using and why~
 - a Why is this component used here~
- a Be confident during viva — you don't need to be a genius, just well-prepared and fluent
- a For your lab report/graph
 - a Use AI tools or software (e.g., Desmos, Excel, ChatGPT) to draw neat graphs and plots
 - a Label axes and units correctly, include standard errors if needed
 - a Keep reports well-formatted — presentation makes a difference
- a Submit your work on time — late submissions may lead to mark penalties or reduced evaluation priority
- a Ask doubts politely during the lab — instructors appreciate interaction and may reward participation subconsciously.

Grading & Evaluation

EMO Lab grading is typically based on the following components

- a Weekly Lab Sessions (~60%)
(Includes viva, performance, submission)
- a Final Presentation (~15–20%)
(Group presentation on one assigned experiment)
- a Lab Compre (~15–20%)
(Based on formulas, procedures, and PYQs)

Even if you don't ace every viva, consistent submission and a strong presentation can land you in the A or A– zone.

Presentation Tips

- a Use clean, modern AI-powered templates (e.g., Gamma.app, Canva, etc.)
- a Practice the flow and divide the script equally among teammates
- a Know your experiment well — the instructors ask cross-questions during the presentation

Final Thoughts

EMO Lab is the least stressful and most scoring course of the semester. Unlike theory courses, success here depends on

- a Discipline (submitting on time)
- a Clarity (understanding the objective of each experiment)
- a Confidence (handling viva and presentation)

If you treat this course with care and avoid slacking off on reports or vivas, it will likely be your strongest grade contributor this semester.

Let it be the easy A that balances out tougher courses like Classical Mechanics or ERA — and also an opportunity to build a rapport with the department (helpful for SOPs or recos later on).

Msc. Chemistry

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Physical Chemistry I (CHEM F211) Course

Content

Physical Chemistry I is a core, mathematically-inclined course that forms the foundation of chemical thermodynamics, equilibria, and molecular behavior. It is often the first real test of a chemistry student's ability to blend theory, logic, and mathematical formalism. The topics covered are essential not only for understanding chemical systems but also for higher-level courses like Physical Chemistry II, Statistical Mechanics, Surface Chemistry, and Spectroscopy. This course is not rote-heavy like some organic/inorganic counterparts — it requires consistent problem-solving and logical derivation practice. A significant portion of the course is conceptualized through differential and integral calculus, and students who treat it like physics with chemical context often perform well.

The course is typically divided into the following modules:

1. Kinetic Theory of Gases
 - U Derivation of gas laws from molecular theory
 - U Maxwell-Boltzmann distribution of molecular velocities
 - U Mean free path, collision frequency, transport properties
2. Chemical Thermodynamics
 - U Laws of thermodynamics
 - U Thermodynamic functions: enthalpy, entropy, Helmholtz and Gibbs free energy
 - U Thermodynamic processes (isothermal, adiabatic, etc.)
 - U State and path functions, exact differentials, thermodynamic equations of state
 - U Calculation of ΔG , ΔH , ΔS using state functions
3. Phase Equilibria
 - U Phase diagrams of one- and two-component systems
 - U Clausius-Clapeyron equation
 - U Gibbs phase rule
 - U Ideal and non-ideal systems
4. Chemical Equilibrium
 - U Equilibrium constants (K_p , K_c , K_x)
 - U Gibbs energy and equilibrium relations
 - U van't Hoff equation (temperature dependence)
 - U Coupled reactions, Le Chatelier's principle
5. Colligative Properties and Solutions
 - U Vapor pressure lowering, boiling point elevation, freezing point depression, osmotic pressure
 - U Raoult's law, Henry's law, ideal vs non-ideal solutions
 - U Derivation of colligative property relations
6. Additional Topics
 - U Surface tension, adsorption, partial molar properties, residual properties (if covered)

What Makes This Course Challenging

- U Heavy use of calculus and partial derivatives
- U Conceptual shift from descriptive chemistry to quantitative interpretation
- U Exams often include derivation-heavy long-form questions and multi-step numericals
- U Questions are not always direct — applying core ideas to unfamiliar setups is common

Tips to Succeed

- U Revise basic calculus and thermodynamic math early (partial derivatives, chain rule, logarithmic properties)
- U Focus on why each equation works, not just how to plug values into it
- U Maintain a derivation notebook — write out every derivation at least 2–3 times with annotations
- U Practice numericals involving multiple steps — especially with state functions and equilibrium
- U Attend every lecture — professors often emphasize what they'll ask in their verbal explanations
- U Clarify units and assumptions — thermodynamics is full of traps with units and constants
- U For phase rule problems, diagram clarity matters — label axes, curves, and key points.

Evaluative Components

- U Tutorials/Short Quizzes – 20–25%
 - U Often derivation-based or numerical; conducted during class or tutorial
- U Midsem Exam – 25–30%
 - U Mix of theory and numericals, can have long derivation questions
- U Comprehensive Exam – 40–50%
 - U Derivation-heavy, often includes application-type numericals and graphical interpretation
- U Assignments (if any) – Typically optional or bonus

Grading & Course Average

- U Moderate to difficult grading, with marks often widely spread
- U A Grade: Requires consistency across all components and strong comprehension
- U A–/B+: Possible with good internal scores (quizzes + midsem), even if comprehension is tricky
- U Course Average: Usually around 55–60, but fluctuates with instructor difficulty and paper structure

Final Thoughts

CHEM F211 introduces you to the language of energy, equilibrium, and systems — not just within chemistry but across physics and engineering. It trains you to think precisely, derive logically, and approach chemical behavior from a first-principles perspective. Success depends on consistent effort, not last-minute cramming. Treat this course as mathematical chemistry, and it will not only become manageable but even enjoyable. Done right, it becomes one of the most empowering courses in the chemistry curriculum.

Msc. Chemistry

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Organic Chemistry I (CHEM F212) Course

Content

Organic Chemistry I is a foundational course in synthetic and mechanistic organic chemistry, designed to develop a deep understanding of carbon-based reaction pathways, stereochemistry, and reactivity principles. It is often viewed as a shift from memorization-heavy high school organic chemistry to a more logic-driven and mechanistically rigorous subject. This course covers the basic structure, bonding, stereochemistry, nomenclature, and reactivity of organic molecules, along with the theoretical frameworks used to explain these properties. Rather than just learning “what reacts,” the course emphasizes why reactions happen, using arrow-pushing mechanisms and molecular orbital theory.

The course is typically divided into the following major modules:

1. Structure and Bonding
 - T Hybridization, orbital overlap, resonance, aromaticity
 - T Molecular orbitals, HOMO-LUMO interaction
 - T Acidity and basicity in organic molecules
2. Stereochemistry
 - T Chirality, enantiomers, diastereomers
 - T R/S and E/Z nomenclature
 - T Optical activity, meso compounds, conformational analysis
3. Reaction Mechanisms and Intermediates
 - T SN1, SN2, E1, E2 mechanism
 - T Carbocation, carbanion, radical and carbene stability
 - T Substitution vs elimination conditions
4. Functional Group Chemistry
 - T Alkanes, alkenes, alkynes — structure and reaction
 - T Alcohols, ethers, epoxides — synthesis and reactivity
 - T Halides, acids, esters — basic transformations
5. Reaction Pathways
 - T Electrophilic and nucleophilic addition and substitution
 - T Rearrangement reaction
 - T Regioselectivity and stereoselectivity principles

The course relies heavily on arrow-pushing formalism and mechanistic logic, so understanding electron movement is key to predicting reactivity and product formation.

What Makes This Course Challenging?

- T Huge volume of content — many reactions to understand
- T Requires remembering mechanisms, reagents, and conditions
- T Some reactions may look similar but follow very different mechanisms
- T Stereochemistry questions often appear in unfamiliar contexts

Tips to Succeed

- T Focus on mechanistic understanding, not just reaction memorization
- T Practice drawing reaction pathways step-by-step — arrow pushing must be precise

- T Maintain a reaction sheet with conditions, reagents, and example
- T Use molecular models or software to visualize stereochemistry and conformation
- T Practice past-year papers to see how known reactions are twisted in exam
- T Attend lectures — professors often highlight what will (or won't) be tested

Evaluative Component±

- T Tutorial Quizzes / Class Tests – 20–25@
 - T Often very direct, sometimes MCQ-base
- T Midsem Exam – 25–30@
 - T Structure-based, mechanistic questions, may include reaction prediction~
- T Comprehensive Exam – 40–50@
 - T Mix of short answer, mechanism writing, stereochemistry, and pathway design~
- T Assignments – Occasionally given for bonus/compulsory topics

Grading & Course Averag

- T Course Average: Generally moderate (~55–60%C)
- T A Grade: Requires high accuracy in mechanism and stereo question
- T B+/A–: Possible with good internal + decent compre performanc
- T Mistakes in arrow-pushing, incomplete mechanisms, or wrong stereochemistry cost marks

Final Thoughts

Organic Chemistry I is not about memorizing 500 reactions. It's about developing a chemist's logic: how molecules behave, why they react, and how to predict outcomes. If you engage with the material regularly and understand the language of organic mechanisms, this course can become intuitive, even enjoyable. It also lays the essential groundwork for Organic Chemistry II and many electives — so building a strong base here is crucial.

Msc. Chemistry

ULTIMATE

Sophomore Guide

Physical Chemistry II (CHEM

F213) Course Content

Soumya Yadav (Msc Math + BE CS)

Physical Chemistry II builds directly upon PC1 (CHEM F211), but shifts focus from classical thermodynamics and equilibria to chemical kinetics, quantum chemistry, spectroscopy, and statistical mechanics. It is a conceptually rich and mathematically intensive course, requiring consistent effort and a solid grasp of both theory and calculus.

This course is especially important for students pursuing theoretical or computational chemistry, spectroscopy, or physical techniques in research. It integrates both classical and quantum perspectives to explain how molecules move, interact, absorb light, and change over time.

The course is generally organized into the following modules:

1. Chemical Kinetic
 - [Rate laws, order of reactions, integrated rate equation
 - [Temperature dependence (Arrhenius and Eyring equations)
 - [Reaction mechanisms and steady-state approximation
 - [Chain and consecutive reaction
 - [Catalysis and enzyme kinetic
 - [Experimental techniques and data interpretation
2. Quantum Chemistry
 - [Failures of classical mechanics and origin of quantum theory
 - [Schrödinger equation and its application to model systems
 - [Particle in a box, harmonic oscillator, rigid rotor, hydrogen atom
 - [Quantum numbers, orbitals, nodes, angular momentum
 - [Operator algebra and postulates of quantum mechanics
3. Molecular Spectroscopy
 - [Interaction of radiation with matter
 - [Types of spectra: rotational, vibrational, electronic
 - [Selection rules, intensity, and transition
 - [IR, UV-Vis, microwave, and Raman spectroscopy (introductory level)
 - [Franck-Condon principle, energy level diagrams
4. Statistical Thermodynamics (Brief Intro)
 - [Ensemble concept, Boltzmann distribution
 - [Partition functions and thermodynamic quantities
 - [Relation to macroscopic thermodynamics

What Makes This Course Challenging

- [Heavy math requirement — calculus, differential equations, eigenvalue problem
- [Requires physical visualization of abstract concepts like wavefunctions and orbitals
- [Many derivations (especially in quantum section) are long and non-intuitive
- [Spectroscopy content can be dense and detail-oriented
- [Kinetics questions often demand logical breakdown of multistep reactions

Tips to Succeed

- [Revise your PC1 fundamentals, especially thermodynamic principles and calculus tools
- [Make a habit of solving quantum derivations step-by-step — no shortcuts here
- [Create a formula sheet and memorize key energy expressions and operators
- [For kinetics, practice decomposing mechanisms and applying steady-state approximation

- [Use visual tools to understand orbitals, energy levels, and spectra
- [Don't just memorize spectra types — know what physical process each corresponds to
- [Consistency > cramming — every module builds heavily on the previous

Evaluative Component

- [Tutorial Quizzes / Assignments – 20–25C
 - [Numerical or derivation-based; conceptual MCQs also possible
- [Midsem Exam – 25–30C
 - [Balanced between kinetics and quantum
- [Comprehensive Exam – 40–50C
 - [Includes spectroscopy, quantum, and kinetics; often multi-part problem
- [Assignments – Given for practice, sometimes graded

Grading & Course Average

- [Course Average: Moderate (usually ~50–55%)
- [A Grade: Requires strong performance in both internal and comprehensive — no weak module
- [A–/B+: Possible with solid internal + decent grasp of quantum
- [Performance typically drops in quantum-heavy semesters if not revised regularly

Final Thoughts

CHEM F213 marks the transition into real physical chemistry thinking. It's no longer about isolated formulas — you'll need to integrate physics, math, and chemistry into a single analytical framework. That can feel overwhelming at first, but also makes it one of the most powerful and rewarding courses if approached steadily. With discipline and a problem-solving mindset, this course builds the kind of deep understanding that defines a chemist.

Msc. Chemistry

ULTIMATE

Sophomore Guide

Inorganic Chemistry I (CHEM

F214) Course Content

Soumya Yadav (Msc Math + BE CS)

Inorganic Chemistry I introduces students to the fundamentals of atomic structure, periodicity, bonding theories, and molecular geometry, forming the conceptual framework for modern inorganic chemistry. The course is heavily conceptual and theory based, with minimal numerical content, and relies significantly on class notes and consistent lecture attendance. This course lays the foundation for advanced topics like coordination chemistry, organometallics, and solid-state chemistry, and is structured to sharpen both factual recall and logical understanding of inorganic trends and behaviors.

The key modules generally include:

1. Atomic Structure
 - \ Quantum numbers, radial and angular functions
 - \ Aufbau principle, electronic configuration anomalies
 - \ Penetration, shielding, and Slater's rule
 - \ Effective nuclear charge (Z_{eff}), ionization energy trends
2. Periodicity and Chemical Properties
 - \ Trends across groups and periods
 - \ Metallic and non-metallic character
 - \ Electron affinity, electronegativity scales (Pauling, Mulliken)
 - \ Inert pair effect, diagonal relationships
3. Ionic Bonding and Lattice Energy
 - \ Ionic character, Born-Haber cycle, Kapustinskii equation
 - \ Radius ratio rule and its application to ionic structures
 - \ Polarizing power and Fajans' rules
4. Covalent Bonding and VSEPR Theory
 - \ Lewis structures, resonance, formal charge
 - \ Valence bond theory and hybridization
 - \ Molecular orbital theory (homonuclear and simple heteronuclear diatomics)
 - \ VSEPR theory and geometry prediction
5. Acids, Bases, and Oxidation States
 - \ Hard and soft acids and bases (HSAB)
 - \ Redox reactions, standard electrode potential
 - \ Oxidation number determination and redox titrations
6. Special Topics (May Vary)
 - \ Introduction to solid state (packing, unit cells)
 - \ Basic symmetry elements and point groups (if time permits)

What Makes This Course Unique

- \ Conceptual, not numerical — clarity of logic matters more than calculation
- \ High reliance on lecture delivery and note-taking — many quiz and exam questions come word-for-word from class
- \ Some memorization is unavoidable, but must be backed by understanding periodic trends and logic
- \ Definitions, trends, exceptions, and reason-based MCQs are common evaluation formats

Tips to Succeed

- \ Attend every lecture — even simple quizzes can be based on a random sentence from class
- \ Maintain clean, highlighted notes with key terms, exceptions, and professor's explanation
- \ Learn definitions word-for-word where required (e.g., HSAB theory) — precision matters

- \ For MO diagrams, practice drawing and labeling each one at least a few times before exam¶
- \ Focus on how periodic trends interconnect, rather than memorizing them in isolation
- \ Don't ignore small topics — basic questions are often asked from corners of the syllabus¶
- \ Use NCERT + class notes + lecture handouts for concise revision

Evaluative Component±

- \ Quizzes / Class Tests – 10–15>
 - \ Usually in-class, short, attendance-based, and easy to score if present
- \ Midsem Exam – 25–30>
 - \ Mostly conceptual, some short answer and MCQ¶
- \ Comprehensive Exam – 40–50>
 - \ More reasoning-based questions, occasionally lengthy descriptive answers¶
- \ Assignments – Rare, unless the instructor chooses to add a component

Grading & Course Average¶

- \ Course Average: Usually around 60–65>
- \ A Grade: Highly dependent on regular class attendance and clean notes¶
- \ Quiz weightage can make or break internal performance, especially if missed
- \ Grading is generally generous, but inconsistent performers may fall quickly

Final Thoughts

CHEM F214 rewards regularity, attentiveness, and careful note-making. It's not conceptually heavy in terms of problem-solving, but demands depth of understanding, memorization of exceptions, and sharp focus during class. For those who attend every lecture and revise class notes regularly, this is one of the least stressful and most scoring theory-heavy courses in the chemistry curriculum.

Msc. Biology

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Biochemistry (BIO F211)

Course Content

Biochemistry (BIO F211) is widely considered the most rigorous course in the 2-1 semester. It is a demanding subject that tests both memory and analytical thinking, with an emphasis on understanding biological mechanisms at a molecular level. The course spans a broad range of biochemical concepts including enzymatic activity, metabolic pathways, and biochemical energetics.

The portion taught by Deepa Ma'am, focuses on conceptual foundations and introduces key topics in a structured manner. Her classes are dense and require undivided attention. She frequently incorporates live demonstrations and relevant videos that help connect theoretical concepts with real-world applications. Tests follow many of these sessions, so staying punctual and attentive is crucial.

The portion taught by Shibashish Sir covers practical and numerical content such as calculations related to enzyme kinetics and biochemical constants. His quizzes are usually more scoring than the main exams, but he also conducts surprise assessments and gives out take-home assignments. Understanding the logic behind numericals is essential, as rote learning will not help during exams — especially in open-book formats.

Tips to Succeed

- T Attend every class — the lecture content often deviates from the textbook, and slides may omit exam relevant material
- T Keep detailed and organized class notes — they become your most valuable asset during revision and open-book exams
- T Understand concepts thoroughly rather than just memorizing facts. Exam questions often combine multiple topics
- T Numerical problems require conceptual clarity — regular practice is key
- T Don't rely solely on slides or the book; you need a synthesis of class content, numerical aptitude, and lab skills.

Evaluative Component

- T Mid Sem – 60 mark
 - T Compre – 80 mark
 - T Lab Components (including Lab Compre) – 60 mark
 - T Quizzes & Other Components – 60 marks
- (All components except midsem and compre were conducted for 120 marks and scaled to 60, with no buffer every mark counts.)

Grading & Course Average¶

- T Average: ~60 mark
- T B Range: Average + 20 to 3

T A– Range: Average + 40 to 4
T A Range: Average + 50 and above

Final Thoughts

Biochemistry requires a sustained, disciplined approach. Scoring well isn't about last-minute cramming — it's about steadily building understanding throughout the semester. Attend all classes, master the numericals, and stay on top of lab work. If you're consistent in your efforts and maintain strong notes, you'll be well positioned to secure a top grade in what is arguably the toughest course of the semester.

Msc. Biology

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Microbiology (BIO F212)

Course Content

Microbiology (BIO F212) is a relatively manageable and scoring course in the 2-1 semester, particularly during the first half. It introduces students to the fundamentals of microorganisms, their classifications, physiology, and roles in health, disease, and environmental processes. The course is divided into two distinct teaching styles across the semester, with different challenges and opportunities in each phase.

The pre-midsem portion is conducted by Soumitra Sir, whose lively and engaging teaching style makes the subject approachable and even enjoyable. His explanations are intuitive and focus on building a solid conceptual foundation. He conducts surprise tests with reasonable weightage, so regular attendance and attentiveness during lectures can give you an early academic edge.

The post-midsem portion is handled by Prabhat Sir, who teaches in a more steady and structured manner.

While his delivery is clear, it can feel monotonous to some students, which often leads to decreased attention and performance. This phase introduces more intricate and detailed topics, and consistency becomes even more important. Like his counterpart, Prabhat Sir may also conduct unannounced evaluations, making continued focus essential even after the midsem.

Slides are provided and generally sufficient for theory preparation, but well-maintained class notes are invaluable for revision and clarification. They help fill the gaps and are especially useful for last-minute review before exams.

The lab component is a critical part of the course, with significant weightage. It includes evaluations, practical sessions, and a lab comprehensive — all of which are important for your final grade. Labs require punctuality, preparedness, and cooperation with your RS and faculty. Slacking off here can result in a significant drop in your overall performance.

Tips to Succeed

G Pay close attention to Soumitra Sir's early lectures — they often hint at what will appear in exams_ G Stay consistent after the midsem; don't let the perceived monotony of lectures reduce your effort_ G Keep clean, complete class notes — they serve as your most useful resource during exams_ G Be prepared for surprise quizzes and small evaluations throughout the course_ G Take labs seriously — from procedures to submissions, everything is evaluated.

Evaluative Component

G Mid Sem – 40 mark
G Compre – 70 mark
G Lab Components (incl. Lab Compre) – 60 mark
G Quizzes & Other Components – 30 marks

Grading & Course Average

G Grading is generous compared to BIO F211_
G B Range: Average + 1>
G A– Range: Average + 25 to 3>
G A Range: Average + 35 and above

Final Thoughts

Microbiology offers a great chance to balance your GPA if approached smartly. Build a strong foundation early with Soumitra Sir's lectures and stay disciplined during Prabhat Sir's portion. Focused effort in the labs, consistent class attendance, and clean notes can set you up for an A. With strategic preparation and steady performance across all components, it's very

possible to do well in this course without being overwhelmed.

Msc. Biology

ULTIMATE

Sophomore Guide

Cell Biology (BIO

F213) Course Content

Soumya Yadav (Msc Math + BE CS)

Cell Biology (BIO F213) is often regarded as the most intellectually stimulating course of the 2-1 semester. While it starts off light and concept-driven, the content quickly scales into deeper analytical and interconnected topics. It covers the structure and function of cellular organelles, membrane dynamics, cell signaling, and cell cycle control — all requiring both factual understanding and logical application.

The pre-midsem portion is taught by Bala Sir, who delivers content in a clear, engaging, and student friendly manner. His classes are interactive, and he emphasizes logical thinking over memorization. Most quizzes are announced and predictable, but smaller surprise assessments do occur, making regular attendance beneficial not just for marks, but for building intuition about the subject.

The post-midsem portion is handled by Shashi Sir, who brings deep subject knowledge but is known for a teaching style that can be hard to follow. His tests are almost always unannounced, and the lectures — though dense — help isolate the central themes often targeted in the compre. Understanding his teaching requires consistent class presence and effort to connect the dots yourself.

Although slides are uploaded, they are often incomplete or insufficient — particularly for open-book exams that demand interpretation, integration of concepts, and application across topics. Hence, your own class notes become your most critical tool for success, especially during revision.

A unique feature of this course is its project work, which involves reviewing research papers and presenting findings. While not highly demanding in terms of time, it requires genuine effort and critical thinking to secure good marks and can make a difference in tight grading margins. The reference book for this course is highly recommended. It is well-written, insightful, and extremely helpful both for conceptual understanding and as a reference during open-book exams. It reinforces class content and fills in any gaps from missed or unclear lectures.

Tips to Succeed

- Engage actively with Bala Sir's classes — his style makes complex ideas accessible early on]
- Persist through Shashi Sir's lectures — they're tougher, but key exam content is often hidden in his discussions]
- Use the reference book to enhance your understanding and build confidence in answering open-book exams]
- Maintain structured class notes — they're your lifeline in an exam that tests logic and synthesis]
- Take the project seriously, as it's not just free marks — the evaluation reflects your depth of engagement.

Evaluative Component

- Mid Sem – 50 mark
- Compre – 70 mark
- Quizzes & Assignments – 60 mark
- Project Work – 20 marks

Grading & Course Average

- Grading is fair, leaning toward generous for consistent performers]
- B Range: Average + 25 to 3<
- A– Range: Average + 35 to 4<
- A Range: Average + 42 to 45 and above

Final Thoughts

Cell Biology rewards curiosity, logical reasoning, and a conceptual approach. Rather than just memorizing slides, focus on understanding how cellular systems work together. With consistent attendance, solid note-taking, and smart use of the reference book, you can approach the open-book exams with confidence. If you give this course thoughtful attention throughout the semester, it can become one of the most rewarding academic experiences in 2-1.

Msc. Biology

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Integrated Biology (BIO F214) Course

Content

Integrated Biology (BIO F214) is a concept-heavy and interdisciplinary course that begins with a relatively light, intuitive introduction but builds toward more analytical and interlinked topics as the semester progresses. It aims to bridge various biological domains and focuses on understanding processes across molecular, cellular, and organismal scales. The course is known for requiring critical thinking, especially in later modules.

The pre-midsem portion is taught by Pankaj Sir, whose calm and student-friendly teaching style makes lectures pleasant and easy to follow. His explanations are clear, and he often encourages discussions that help clarify complex ideas. He seldom conducts unannounced evaluations, which gives you space to focus on comprehension rather than test pressure. Attending classes regularly during this phase helps lay a strong foundation.

The post-midsem portion is covered by Meghna Tare Ma'am, who moves at a faster pace. Her classes are more tightly packed, and she often highlights exactly what to focus on for exams — making her lectures extremely valuable for exam preparation. While she may conduct surprise low-weight quizzes, the major evaluations are always announced, with a clearly defined syllabus.

The BITS-custom edition reference book is well-written and conceptually strong but isn't ideal for quick revision. It's best used to deepen understanding after lectures rather than as a cramming tool. For exam preparation, especially the open-book components, your class notes are by far the most important asset — they are tailored to the exam structure and include insights not found in slides or the textbook.

The course has no lab or project component, which can lighten the workload compared to other BIO courses. However, this means the quizzes, midsem, and compre carry all the weight, so consistent performance is critical.

Tips to Succeed

- Attend Pankaj Sir's classes actively — they're paced to help you absorb the core ideas
- Pay close attention to Meghna Ma'am's exam guidance — it's usually spot on
- Build comprehensive class notes — especially valuable in open-book exams that test logic and synthesis
- Use the textbook selectively for deeper insight, not for last-minute preparation
- Don't neglect smaller quizzes — they matter in the overall grade calculation.

Evaluative Component

- Mid Sem – 60 mark
- Compre – 90 mark
- Quizzes & Other Components – 50 marks

Grading & Course Average

- Grading is balanced and rewards consistent, logical work
- B Range: Average + 2@
- A– Range: Average + 35 to 4@
- A Range: Average + 45 and above

Final Thoughts

Integrated Biology is a balanced course that gives back as much as you put in. It doesn't demand excessive memorization, but it does require a logical, structured approach to learning. By attending lectures, focusing on class notes, and maintaining steady performance throughout the semester, you can comfortably secure a strong grade. This course is especially rewarding for students who enjoy making conceptual connections across different areas of biology.

B. Pharma

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Process Engineering (BITS F219) Course

Content

Process Engineering is widely considered the most approachable and high-scoring CDC in the 2-1 Pharma curriculum. Taught by Prof. Anil Jindal, the course introduces students to fundamental pharmaceutical industrial processes, focusing on both conceptual clarity and simple numericals relevant to real-world manufacturing. What makes this course especially student-friendly is how closely aligned the exam questions are with the classroom teaching.

Prof. Jindal is known for his clear, direct, and highly structured lectures. If you attend classes attentively and write down

everything he says, you're already halfway to an A. In fact, nearly 80% of both the open book exam and quizzes directly reflect the discussions held in class. Often, even the numerical problems in the quizzes and finals use the exact same values discussed on the board — something that rarely happens in other CDCs.

While slides are provided, they do not cover every detail. Key explanations, shortcut methods, and numerical tricks are usually only discussed verbally. That's why maintaining well-organized, complete handwritten notes is not just helpful — it's essential. You won't be able to rely solely on the slides during exams, especially if you've skipped classes.

The course had a balanced assessment structure: one subjective quiz and the rest as MCQ-based quizzes. All quizzes were pre-announced and came with a reasonable buffer, meaning there was a healthy margin for scoring well with a bit of preparation. The open-book compre exam was extremely straightforward — the entire question paper was made up of examples and topics explicitly covered in class, with no surprises or twists.

The lab component, while relatively simple in terms of experiment complexity, was evaluated strictly. There were no lab buffers, and RS were strict about punctuality, procedure adherence, and lab conduct. The experiments were not intellectually difficult but required your presence and attention. Skipping labs or underperforming in them could cost you marks in an otherwise high-scoring course.

Tips to Succeed~

- C Attend every single class — the exam paper is a mirror of what's taught in the classroomZ
- C Prioritize making thorough handwritten notes during lectures — they are your primary revision toolZ
- C Practice in-class numericals, as they often appear unchanged in quizzes and finalsZ
- C Treat the labs seriously — while easy, they have no slack in terms of attendance or evaluationZ
- C Use lecture slides for a quick recap, but depend on your notes for real understanding and exam prep.

Evaluative Component

- C Quizzes: Mostly MCQs, one subjective (all pre-announced, buffer included?)
- C Comprehensive Exam: Open-book, fully based on classroom contents
- C Lab Work: Simple experiments, no buffer, strict RS

Grading & Course Average

- C Grading is very generous, especially for students who maintained consistent attendance and preparationZ
- C A significant number of students received A or A– gradesZ
- C This course is a prime CGPA booster if handled with basic consistency.

Final Thoughts

Process Engineering is a rare CDC that rewards steady attendance over brute-force cramming. If you're present in class, take good notes, and revise class numericals, you can walk into the open-book exam fully confident. The labs, while basic, demand your presence and cooperation. Overall, this course offers one of the least stressful and most rewarding paths to an A in the semester — don't waste that opportunity by being complacent.

B. Pharma

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Biochemistry (PHA F242) Course

Content

Biochemistry is one of the most demanding and concept-intensive CDCs in the 2-1 Pharma semester, taught by Prof. Richa. The course explores the chemical foundations of biological processes — including metabolic pathways, molecular structures, and enzyme mechanisms — all of which require strong memorization, clarity in reaction flow, and regular revision.

This is not a course you can tackle at the last minute. The syllabus is dense, and topics are layered — understanding one concept often depends on mastery of earlier ones. Many students underestimate the effort required and struggle as the semester progresses.

Prof. Richa has a blackboard-oriented, structure-heavy teaching style. She draws almost every biochemical pathway and molecular structure on the board, expecting students to follow along and redraw everything themselves. Sitting in front, engaging actively during class, and redrawing/revising these structures regularly is absolutely crucial. Simply depending on slides will not suffice — many exam-relevant points are discussed only during class.

The quizzes are quite tough, with class averages often between 1 to 3 out of 10 — especially if you don't revise regularly. The open-book component of the compre exam tends to include twist-based application questions, often derived from previously asked questions (PYQs). Solving all available previous year questions is highly recommended, and issuing the reference book from the library well in advance is a smart move — it helps during open-book exams for structure-based reasoning and enzyme-specific detail.

The lab component is serious and weighty. There are scheduled vivas throughout the semester, and no lab buffer is given. However, the RS were reported to be good and helpful. It's not enough to just perform the experiments — the theory behind each experiment is frequently asked in vivas and sometimes appears in the theory paper too. Many students lose marks by ignoring theory during lab hours.

Tips to Succeed

- Attend lectures actively and draw every structure and pathway with the professor — this builds muscle memory and clarity
- Don't rely on slides — most high-yield information is blackboard-only
- Revise weekly — the content snowballs quickly and can overwhelm you if left for later
- Use the reference book and solve PYQs for open-book exam prep — application-based questions require strong foundation
- For labs, prepare theory as thoroughly as practicals — it's asked directly in vivas and exams
- Don't take quizzes lightly — even though they're hard, they still contribute to your grade and offer early insights into your standing.

Evaluative Component

- Quizzes: Tough, low averages (~1–3/10)
- Comprehensive Exam: Open-book; requires solving PYQs and using reference book
- Lab Work: No buffer, regular vivas, theory is as important as experiment execution

Grading & Course Average

- Grading is tougher than most other CDCs, particularly compared to Process Engineering
- Only 2–3 students received an A in some previous semester
- Performance gaps are wide — small mistakes can significantly affect the grade

Final Thoughts

Biochemistry is a rigorous but rewarding course if approached the right way. It demands discipline, repetition, and detailed understanding rather than rote memorization. Students who draw along in class, revise consistently, solve PYQs, and respect the lab theory component often find themselves in a stronger position, even if the grading is strict. This is not a course to take lightly, but for those willing to put in the effort, it offers a deep and valuable understanding of the molecular workings of life — and a hard-earned A.

B. Pharma

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Pharmaceutical Analysis (PHA 241) Course

Content

Pharmaceutical Analysis is a moderately challenging CDC that covers various foundational topics in physical chemistry, such as the mole concept, ionic equilibrium, pH calculations, acid-base titrations, and other related areas. Taught by Prof. Sandeep Sundriyal, this course focuses on problem-solving and conceptual application — particularly numericals — and requires a methodical, practice-oriented approach to perform well.

The course has a strong numerical focus, especially in the open-book section of the comprehensive exam. Many of these questions involve detailed calculations and require a strong grip over basic physical chemistry concepts. The reference book is extremely useful here — it offers clarity, solved examples, and exam-level problems that align well with what's asked in finals. You are advised to issue the reference book early from the library and use it throughout the semester for both learning and exam prep.

Lecture slides are well-prepared, clear, and highly usable, particularly for the closed-book section of the comprehensive exam. However, class attendance is still crucial. There were multiple unannounced short quizzes throughout the semester, many of which significantly impacted final grades. These were strictly evaluated, often with no buffer and negative marking — making every quiz attempt high-stakes. Missing even one quiz could affect your grade due to tight scoring margins.

The lab component revolves around simple titrations and physical chemistry-based experiments, which are generally easy to execute. However, as with other courses, there were no buffers in labs, and attendance and performance were strictly evaluated. While the procedures themselves aren't complex, accuracy, punctuality, and lab record maintenance are necessary to secure full marks.

Tips to Succeed

- Focus heavily on numericals — the open-book component tests application, not just concepts
- Attend all lectures — unannounced quizzes carry weight and are harshly graded
- Use the reference book extensively — it will help during exam time and builds a strong base
- Slides are sufficient for closed-

book revision, but class explanations often include exam hints
Y D Be consistent with lab work — errors in titration or missed submissions can cost you
Y D Practice problem sets regularly — chemistry here is about application, not just theory.

Evaluative Component

D Quizzes: Frequent, unannounced, strict evaluation with no buffer and negative marking
D Comprehensive Exam: Divided into open-book (numericals) and closed-book (theory from slides; D Lab Work: Titration-based, no buffer, performance and punctuality matter

Grading & Course Averag

D Grading is strict and competitivw
D No leniency in marking — small slip-ups in quizzes or labs can heavily impact the final gradw
D CGPA gain is possible, but only with consistent effort across all components

Final Thoughts

Pharmaceutical Analysis can feel unforgiving if approached casually, but it becomes highly manageable with the right strategy: numerical mastery, full attendance, and a proactive attitude toward surprise quizzes. The reference book is your best friend for the open-book component, and regular practice is key to navigating the calculation-heavy questions. Respect the structure, and this course can become a solid contributor to your academic profile — just don't expect it to hand out marks for free.

B. Pharma

ULTIMATE Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Gen Maths-III (BITS F218) Course

Content

Gen Maths-3 is widely considered the easiest and most CGPA-friendly CDC in the 2-1 Pharma semester. Taught by Prof. B.K. Sharma, the course includes linear algebra, differential equations, vector calculus, and related applied mathematical tools that are foundational for technical disciplines. It's particularly well-received due to its simple explanations, manageable pace, and direct relevance to standard board level math.

The course builds upon some topics from Gen Maths-2 but is significantly easier in terms of difficulty. If you struggled with Gen Maths-2, don't worry — most students find this course far more digestible. In fact, to get a head start, many recommend revisiting the Matrix chapter from Class 11/12 NCERT or YouTube resources, as it forms a solid base for much of the initial content.

The most important part of succeeding in this course is following a sequential learning approach — the content is cumulative. Each week builds on the last, and if you fall behind early, it can snowball. Prof. Sharma explains every concept in detail and walks through problem-solving step-by-step during lectures. He also works out example problems in class, which are highly likely to reappear in quizzes or finals with minimal variation.

Quizzes were mostly unannounced, but buffers were given, making it less punishing than other CDCs. Regular class attendance and problem practice were enough to stay ahead. The questions asked were almost always drawn from in-class examples or simple variations of them, making it easy to prepare if you kept up week by week.

There is no lab component, which further lightens the course load. All evaluations are based purely on theory and problem-solving — no vivas, no experimental reports.

Tips to Succeed

Attend all lectures, and solve every example discussed in class — they often reappear in assessments
Study sequentially — avoid skipping chapters or jumping ahead; the flow matters
Use Class 11/12 Matrix resources to reinforce your fundamentals early on
Do all class-assigned problems and PYQs — they form the backbone of quiz and exam questions
Don't stress about surprise quizzes — the buffer makes them manageable if you're consistent.

Evaluative Component

Quizzes: Unannounced, but buffers provideh
Comprehensive Exam & Midsem: Direct application of classwork problemu
No lab component — purely problem-solving focused

Grading & Course Averag

Grading was excellent, with a large portion of the class securing A / A– gradeu The course is regarded as a strong CGPA booster, especially when combined with even basic preparation

Final Thoughts

Gen Maths-3 is your best opportunity to score high with minimal stress. If you attend lectures, follow the class flow, and consistently solve problems, an A is very much within reach. With no lab pressure and a student-friendly teaching style, this course stands out as the perfect GPA lifter — just don't let its simplicity make you complacent.

Common Courses **ULTIMATE** Sophomore Guide

Soumya Yadav (Msc Math + BE CS)

Mathematics - III (MATH F211) Course

Content

Mathematics III (commonly referred to as M3) is a core mathematics course focusing on differential equations, linear algebra, and Fourier analysis. It's essential for engineering students across branches as it lays the mathematical foundation for advanced applications in signal processing, systems theory, control systems, and PDEs. The course generally includesQ

- a Ordinary Differential Equations (ODEs): First and second-order, homogeneous and non-homogeneousu a Systems of Differential Equation
- a Laplace Transform
- a Fourier Series and Transform
- a Partial Differential Equations (PDEs)
- a Introduction to Linear Algebra: Eigenvalues, eigenvectors, diagonalization (depending on the instructor) Compared to M2 or PNS, M3 is more predictable and easier to plan for, especially if you pick a consistent prep source.

Tips to Succeed

- a Pick one consistent learning source and stick to itQ
 - a Either attend lectures regularly, o
 - a Use Shekhawat Sir's online videos (YouTube) — both are sufficient to cover the entire syllabusu a You can mix and match if needed — for example, attend lectures but refer to videos for tougher topics like PDEs or Fourier Seriesp
- a No surprise tests: Unlike PNS or M2, M3 quizzes are announced well in advance — usually in the first 10–12 days of the semester. You can plan your schedule accordinglyp
- a Solve as many PYQs as possible. They are the most effective preparation tool, especially for quizzes and midsems. Most question types repeat with minor variationsp
- a Go through suggested textbook problems at least once. They may seem repetitive, but they reinforce important solution techniquesp
- a Midsem tends to be easier — try to build a strong score buffer here. The post-midsem portion is conceptually denser and often more abstract (especially PDEs and Fourier)p
- a Focus heavily on quizzes. They are straightforward and scoring opportunities — missing one can hurt your grade significantlyp
- a Time management in tests is important. Even if the content is simple, the paper can be lengthy, especially in the compre.

Grading & Evaluation

- a Tut Tests / Quizzes: Best 2 out of 3, Announced well ahead of time, usually within the first few weeks. Focused on earlier, simpler portions of the syllabus — easy way to boost your averagep
- a Midsem: Often from PYQs + tutorial sheets. Focus on clarity of method and presentationp
- a Compre: More challenging — covers complex portions like PDEs, Fourier series/transforms, and longer questions. General

Grading Trend<

- a Avg + 85 to 90 → Safe for [
 - a Avg + 35 to 40 → Safe for =
 - a Grading pattern is similar to PNS, and professor estimates on cutoffs can be unreliable.
- Example: Someone with avg+95 was told that A begins at avg+97 — which is unlikely. Always take such estimates with a grain of salt.

Final Thoughts

Mathematics III (MATH F211) is predictable, structured, and relatively high-yield. It doesn't throw surprises like M2 or PNS, and with consistent effort, it can become a solid CGP booster.

To succeedQ

- a Stick to one source
- a Stay consistent with PYQs and quizzes
- a Build a strong midsem lead~
- a Don't underestimate the Compre portion

It's a course that rewards smart preparation, not brute memorization, and gives back generously to those who plan ahead.

THANK YOU